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(54) **AUTOMOTIVE PROCESSOR WITH PQK
ENCRYPTION SUPPORT**

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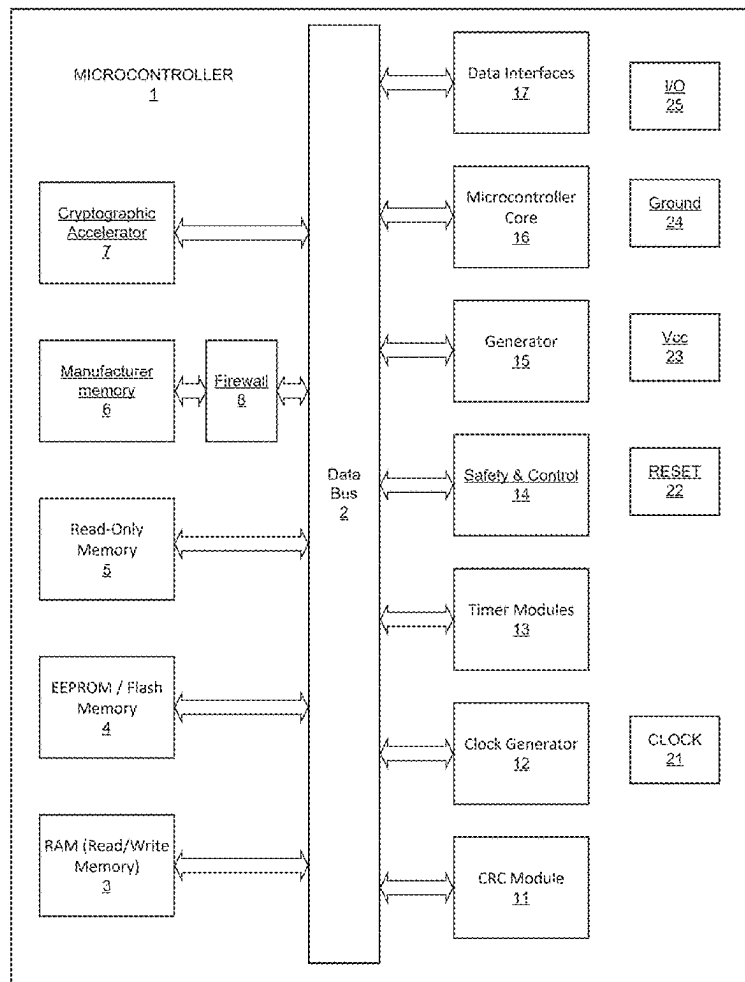
(57) **ABSTRACT**

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A device comprises a quantum random number generator, and further comprises following device parts: a first SPAD diode, a second SPAD diode, an optical fiber which optically couples the first SPAD diode and the second SPAD diode to one another, an amplifier and/or filter, an analog-to-digital converter, a comparator, a time-to-digital converter, and an entropy extraction device converting output values of the time-to-digital converter into first and second values and generates random bits therefrom.

Related U.S. Application Data

(63) Continuation of application No. 18/702,318, filed on Apr. 17, 2024, filed as application No. PCT/EP2022/079826 on Oct. 25, 2022.



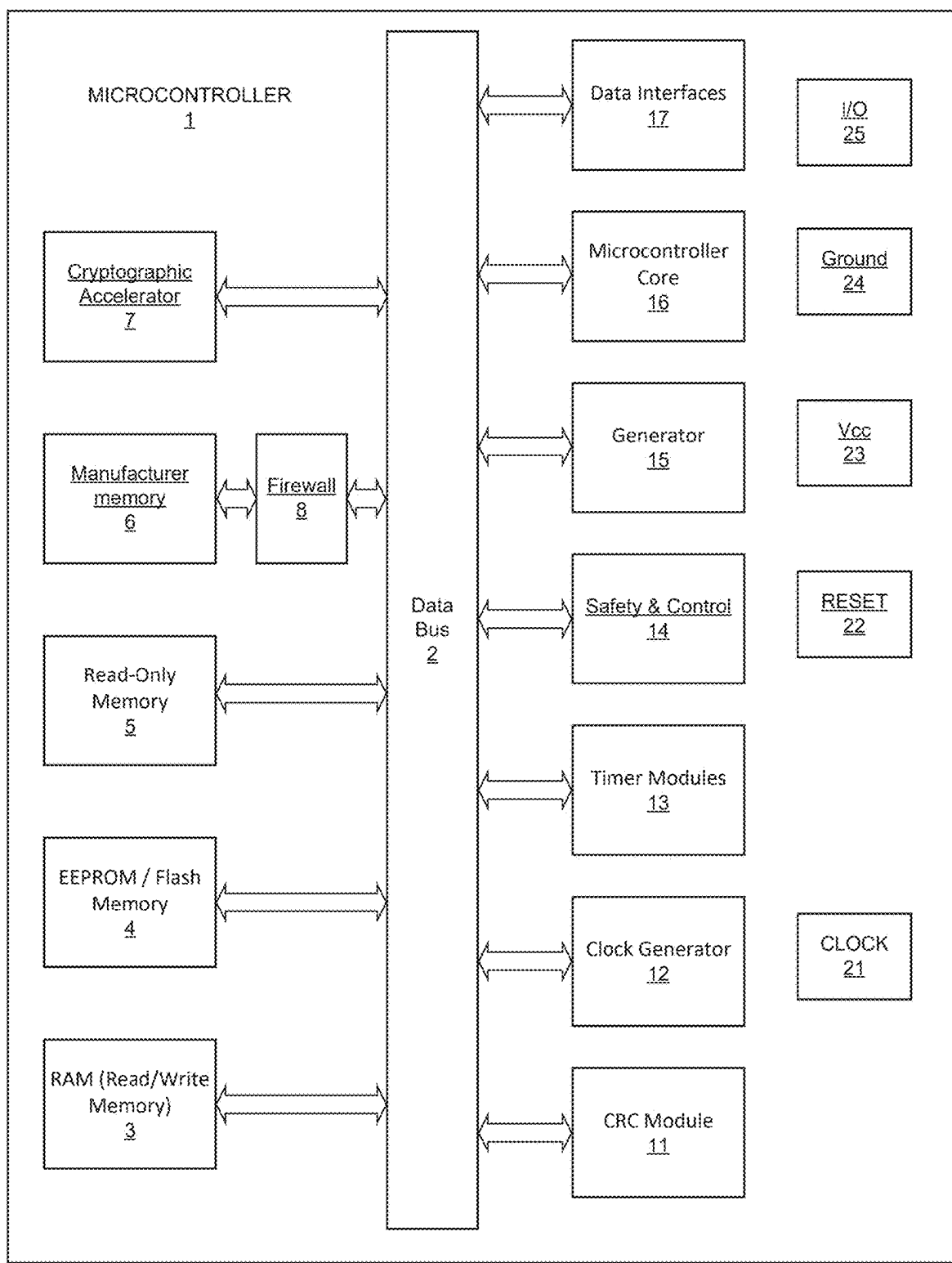


FIG. 1

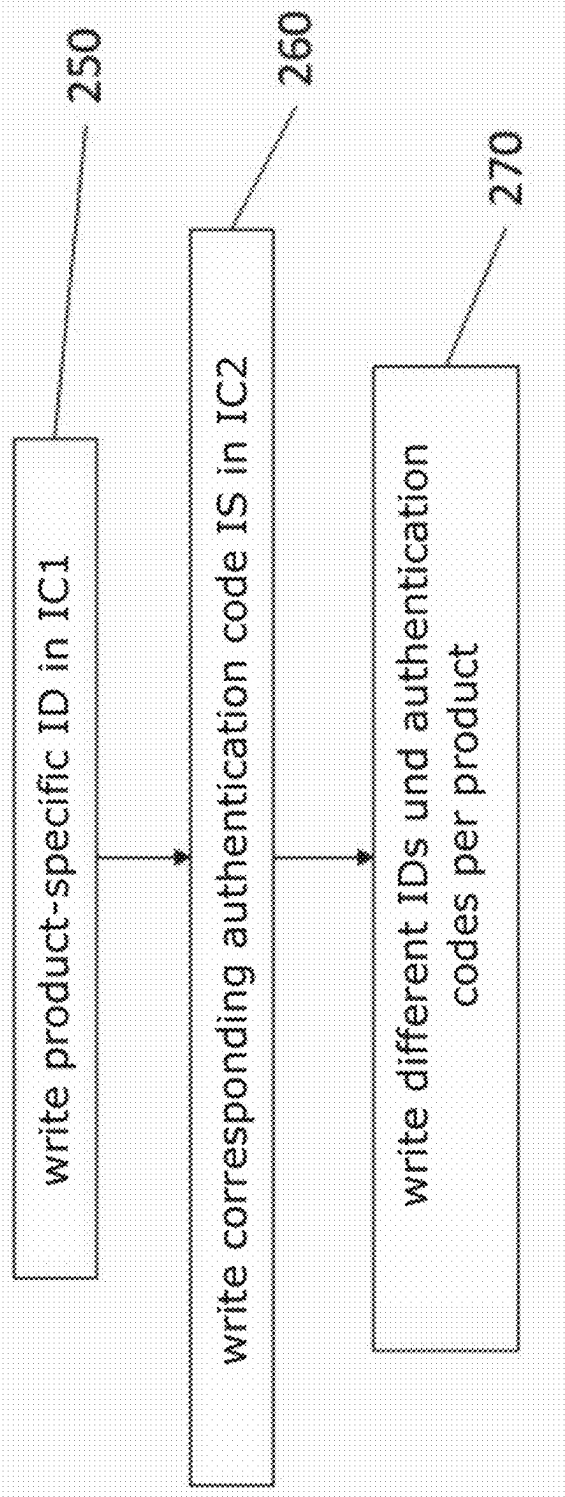


FIG. 2

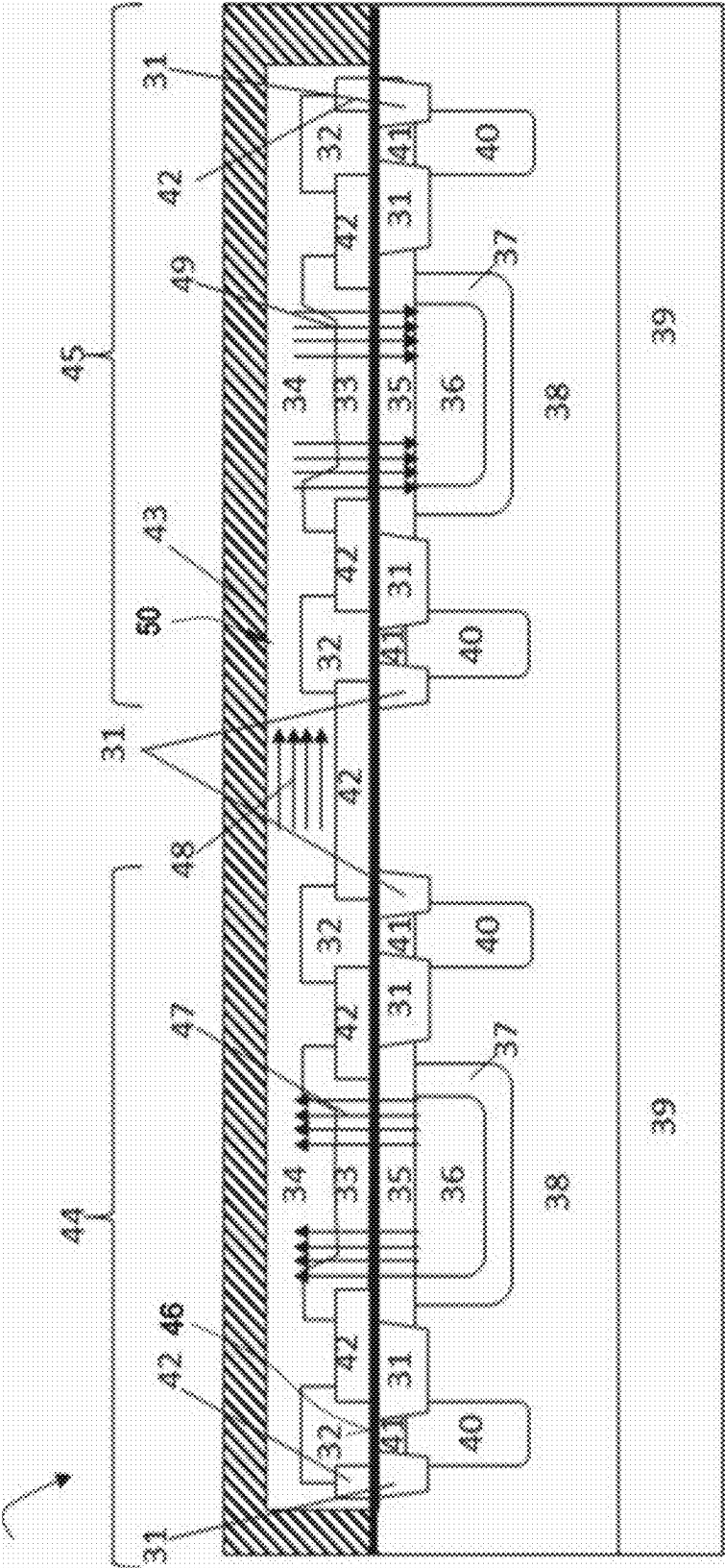


FIG. 4

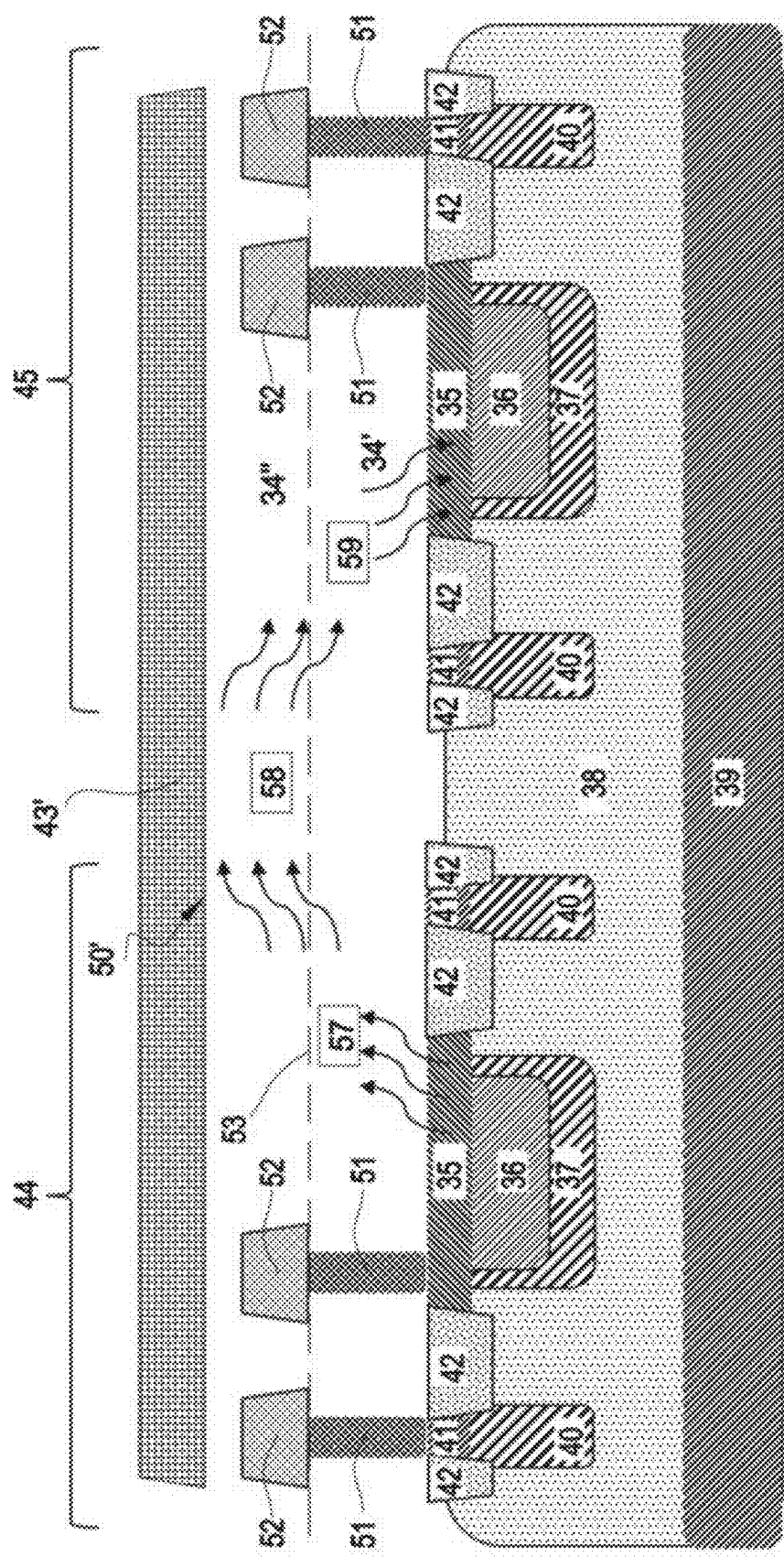


FIG. 5

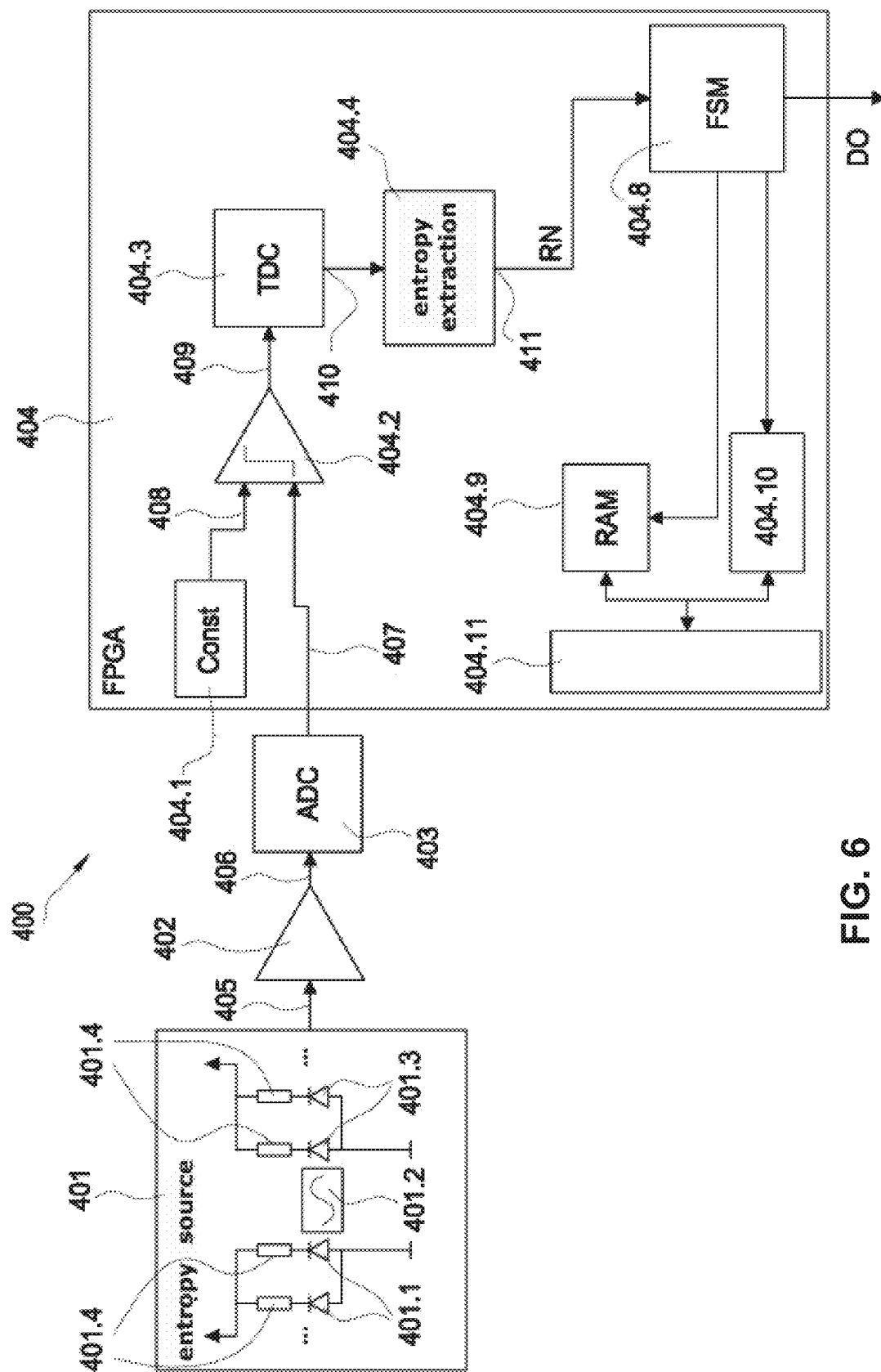
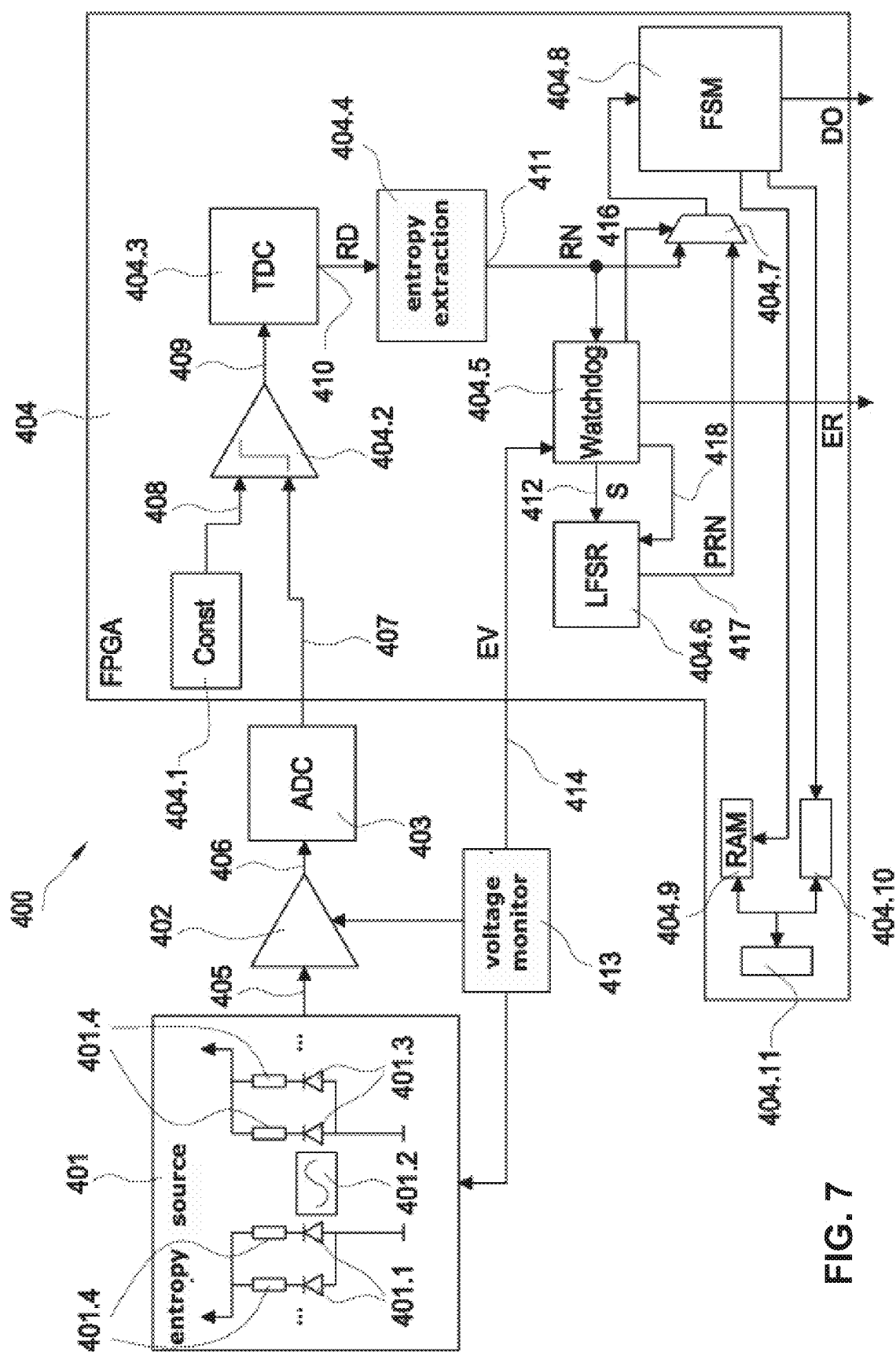


FIG. 6



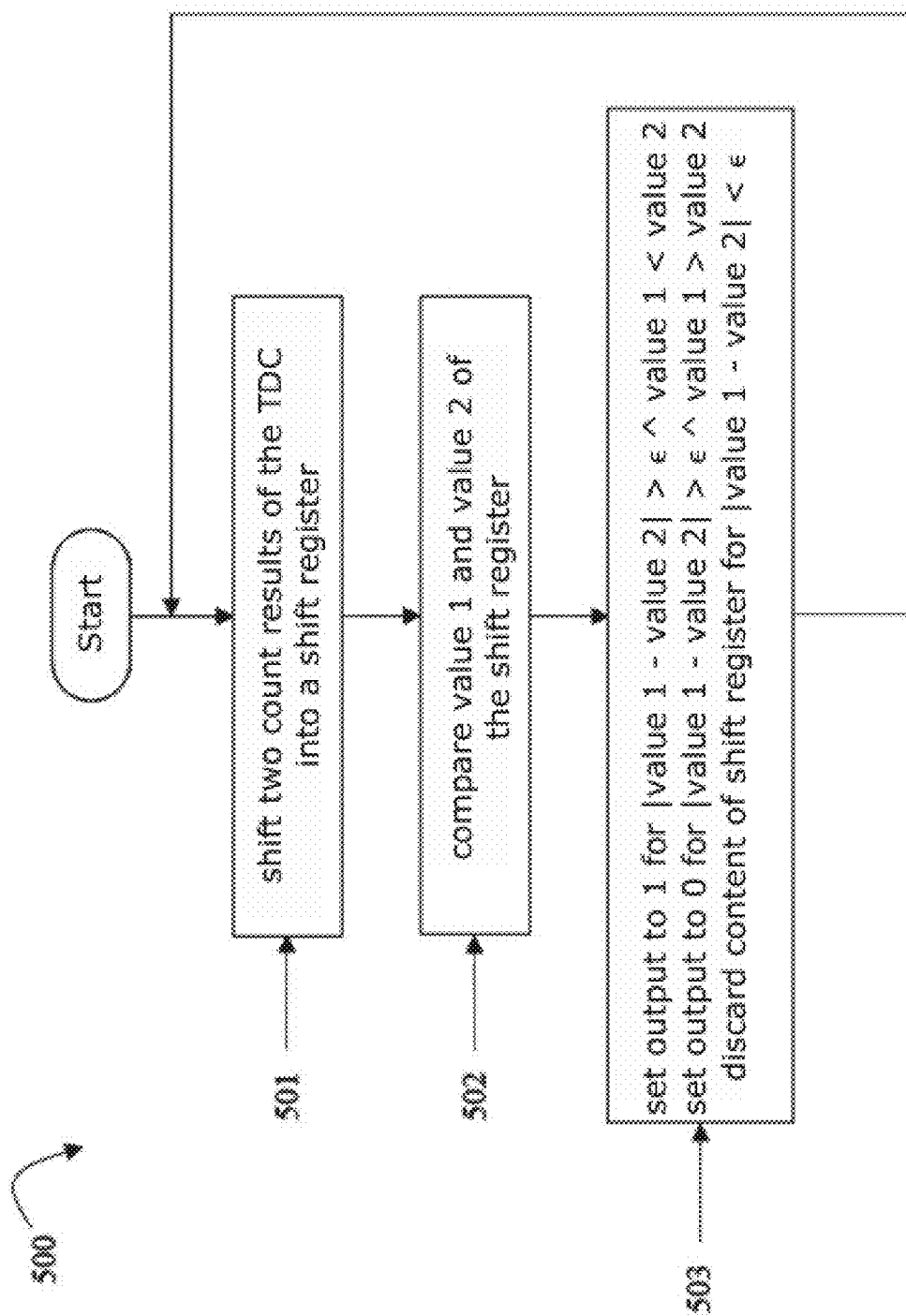


FIG. 8

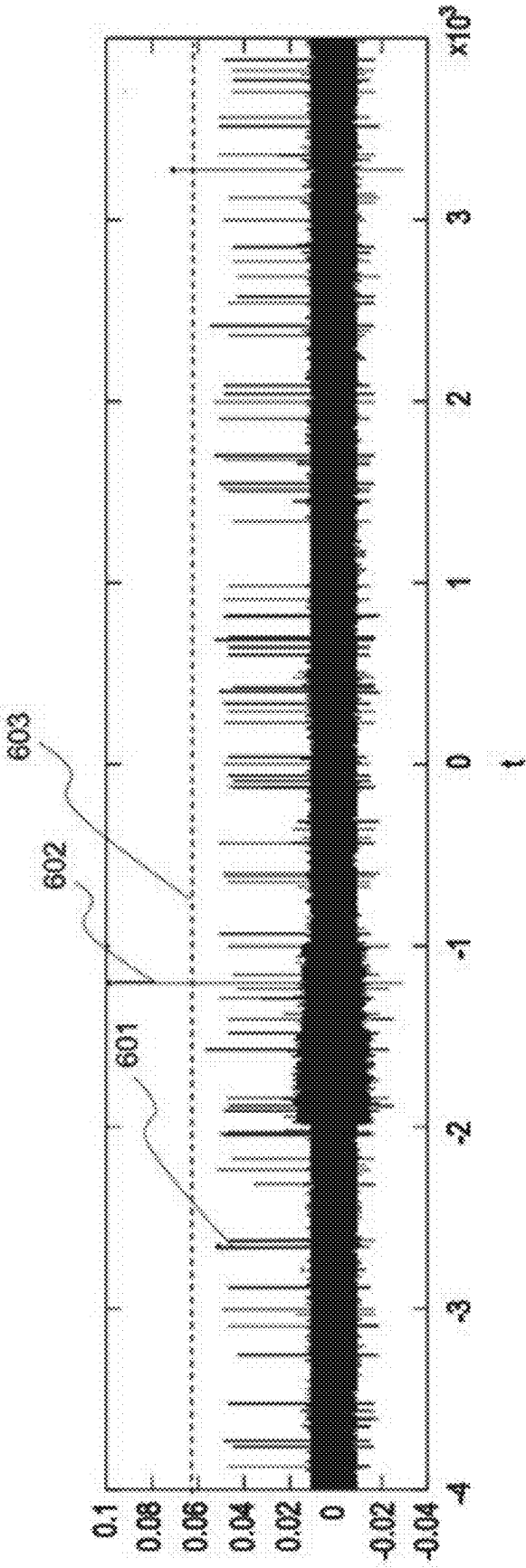
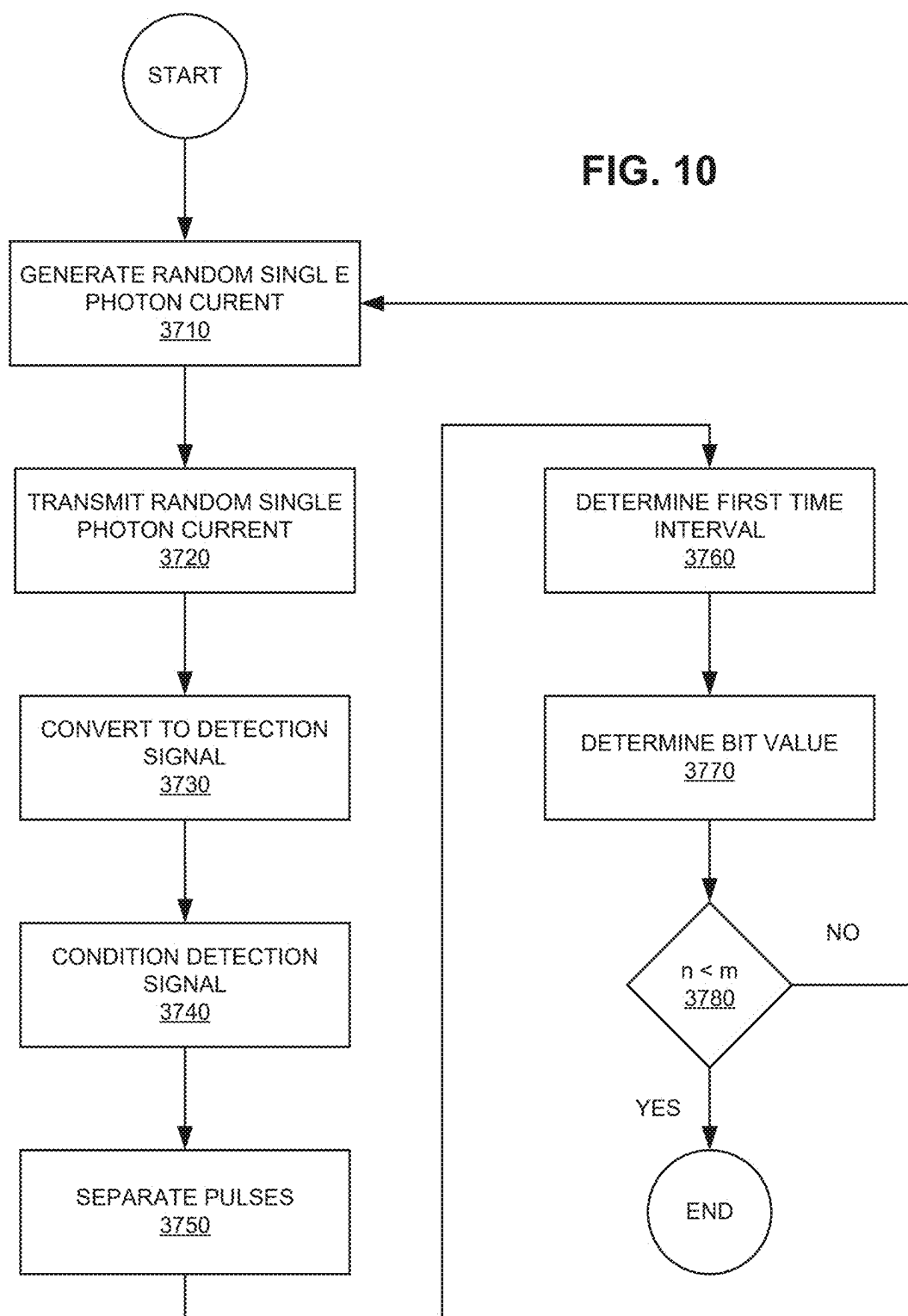


FIG. 9

FIG. 10



AUTOMOTIVE PROCESSOR WITH PQK ENCRYPTION SUPPORT

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is a Continuation of U.S. patent application Ser. No. 18/702,318, filed on Apr. 17, 2024, which is a US National Phase of International Patent Application Number PCT/EP2022/079826, filed on Oct. 25, 2022, claiming priority to the German Patent Applications 10 2021 128 005.2 of 27 Oct. 2021, DE 10 2021 130 107.6 of 18 Nov. 2021, DE 10 2022 110 713.2 of 2 May 2022, DE 10 2022 125 574.3 of 4 Oct. 2022, DE 10 2022 125 617.0 of 5 Oct. 2022 and DE 10 2022 125 768.1 of 6 Oct. 2022, the contents of each of which are hereby incorporated into the subject matter of the present application by reference.

TECHNICAL FIELD

[0002] The disclosure relates to a microcontroller comprising, inter alia, at least one quantum process-based generator for true random numbers (Quantum Random Number Generator: QRNG) as a random number generator, which is applied, for example, for encryption.

BACKGROUND

[0003] The automotive industry and other branches of industry are increasingly exposed to a wide variety of piracy attacks. The counterfeiters copy the spare parts and products of the industrial manufacturers concerned and generally use their brand names. Hacker attacks, e.g. affecting the transmission of data within the products or to and from the product, such as in car-to-car or car-to-x communication or in data communication within the components of a vehicle, are a further risk, which is particularly relevant in autonomous systems and in data communication from outside with such autonomous systems and from these to outside with regard to protection against hacker attacks.

[0004] The entropy properties of currently used random number generators for data encryption in such systems are mostly insufficient. Quantum process-based generators for true random numbers (Quantum Random Number Generator: QRNG) are known in prior art, but they are difficult to integrate or have a poor quantum yield.

[0005] Details of data and signal processing units with microcontrollers and quantum process-based generators for the generation of random numbers are described, for example, in

[0006] U.S. Pat. No. 10,802,800

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[0010] BURRI, S. [et al.]: SPADs for Quantum Random Number Generators and beyond. In: 19th Asia and

South Pacific Design Automation Conference (ASPDAC), 20-23 Jan. 2014, Singapore, IEEE, p. 788-794. DOI: 10.1109/ASPDAC.2014.6742986;

[0011] Fabio Acerbi, Nicola Massari, Leonardo Gasparini, Alessandro Tomasi, Nicola Zorzi, Giorgio Fontana, Lorenzo Pavesi, Alberto Gola in “Structures and Methods for fully-integrated Quantum Random Number Generators”, IEEE Journal of selected topics in quantum electronics, Vol. 26, No. 3, May/June 2020.

SUMMARY

[0012] The object of the disclosure is to provide a device and a method for generating true random numbers on the basis of optical quantum processes.

[0013] The object is achieved with the disclosure proposing a microcontroller for controlling devices in automobiles, wherein the microcontroller is provided with

[0014] a semiconductor substrate,

[0015] memory elements,

[0016] at least one internal bus,

[0017] at least one microcontroller core,

[0018] one or more data interfaces, and

[0019] at least one quantum process-based generator for true random numbers,

[0020] wherein the memory elements are connected to the internal bus,

[0021] wherein the data interface is connected to the internal bus,

[0022] wherein the quantum process-based generator is connected to the internal bus, and

[0023] wherein the microcontroller core is connected to the internal bus,

[0024] wherein the quantum process-based generator generates a random number, optionally at the request of the microcontroller core,

[0025] wherein the quantum process-based generator provides said random number,

[0026] wherein the microcontroller core generates a key using of a program from one or more of its memory elements and using the random number,

[0027] wherein the microcontroller core uses a program from one or more of its memory elements and the key to encrypt and decrypt data which it exchanges with external devices or internally via the data interface,

[0028] wherein the semiconductor substrate integrally comprises the sub-devices of the secure microcontroller mentioned herein,

[0029] wherein the quantum process-based generator comprises at least one first SPAD diode as a light source for an optical quantum signal,

[0030] wherein the quantum process-based generator comprises the at least one second SPAD diode as a photodetector for receiving the optical quantum signal,

[0031] wherein the quantum process-based generator comprises the at least one processing circuit,

[0032] wherein the quantum process-based generator comprises the at least one optical fiber for optically coupling the at least one first SPAD diode to the at least one second SPAD diode,

[0033] wherein the quantum-based generator comprises an operating circuit for supplying the at least one first SPAD diode with electrical energy for emitting light, and

[0034] wherein the quantum-based generator comprises a processing circuit for detecting the signal of the at least one second SPAD diode and for forming a random number from the signal of the at least one second SPAD diode as well as for providing the random number to the microcontroller core.

[0035] A feature of the microcontroller is the quantum process-based generator (quantum process-based entropy source) for the generation of true random numbers. Said generator is based on an optical quantum process, namely on the combination of stimulated and spontaneous photon emissions in at least one second or receiver SPAD diode, respectively. Said receiver SPAD diode is coupled to at least one first or transmitter SPAD diode via an optical waveguide. Both SPAD diodes are operated in Geiger mode.

[0036] A feature of the innovation includes forming the optical fiber as at least part or section of one or more dielectric layers, which is/are part of the metallization stack as insulation layers, which in turn is formed on the semiconductor material in which the first and second SPAD diodes are integrated. Alternatively, the optical fiber can also couple two semiconductor substrates together, in each of which at least one of the first and at least one of the second SPAD diodes are formed. However, it is preferable to design the microcontroller with a semiconductor substrate into which all SPAD diodes are integrated and on the surface of which the optical fiber is arranged.

[0037] Due to the emergence of spontaneous and stimulated photon emissions from the second SPAD diode, its emission signal has extremely narrow pulses (hereinafter referred to as spikes) with different heights. The emission signal thus comprises first spikes that arise as a result of single photon emissions. Second spikes in the emission signal of the second SPAD diode, which are higher than the first spikes, occur when a spontaneous photon emission coincides with a stimulated photon emission. The emergence of such combined photon emissions is highly random, i.e. quantum process-based, which is why the evaluation of the temporal sequence of the emergence of second spikes is now suitable for generating a true random number therefrom.

[0038] In this respect, in an advantageous further development of the disclosure, it is provided that the emission signal of the at least one second SPAD diode has first spikes and second spikes which are larger than the first spikes, wherein the first and the second spikes are each larger than a predefinable reference value, wherein the first spikes arise due to a spontaneous photon emission of the at least one second SPAD diode without a simultaneously caused stimulated photon emission of the at least one second SPAD diode and the second spikes arise due to the simultaneous occurrence of a spontaneous photon emission of the at least one second SPAD diode stimulated by the reception of a photon of the at least one first SPAD diode, and in that the processing circuit of the quantum process-based generator generates a first logical value or a second logical value as a bit for generating the random number on the basis of a comparison of the time intervals of second spikes with a threshold value or on the basis of a comparison of the time intervals of second spikes with one another.

[0039] According to a variant of the disclosure, the true random number is generated such that the processing circuit generates the first logical value or the second logical value on the basis of a comparison of the time interval between two spikes with a threshold value, and in that the processing circuit outputs the first logical value as a bit of the random number if the time interval is smaller than the threshold value, and outputs the second logical value as a bit of the random number if the time interval is greater than the threshold value.

[0040] According to a second variant, the true random number is generated such that the processing circuit generates the first logical value or the second logical value based on a comparison of the time intervals of the second spikes of different pairs of second spikes, and in that the processing circuit outputs the first logical value as a bit for generating the random number if the time interval of the most recently occurred second spike is smaller than the time interval between, for example, the penultimate second spike and the antepenultimate second spike or another pair of previous second spikes, and outputs the second logical value as a bit for generating the random number if the time interval of the most recently occurred second spike is, for example, greater than the time interval between, for example, the penultimate second spike and the antepenultimate second spike or another pair of previous second spikes.

[0041] For both above-described variants of generating the true random number it applies that if equality is detected when comparing the time intervals with the threshold value (variant 1) or with each other (variant 2), this result is discarded for utilization for the random number. Alternatively, the results of the comparisons for equality could also be utilized, but this would then have to be evaluated purely by chance as either the first logical value or the second logical value.

[0042] For the sake of completeness, it should also be mentioned that third spikes can also occur in the emission signal in the second or receiver SPAD diode, which are essentially three times the height of the first spikes. Such events occur when a photon emission, which is stimulated by the reception of a photon of the first SPAD diode, coincides with a spontaneous photon emission from the second SPAD diode, which is in turn stimulated. Advantageously, these "triple spikes" are not used. In this respect, when examining the emission signal of each second SPAD diode, its signal is compared with a predefined reference value range within which it must be located in order to be evaluated for the generation of the true random number.

[0043] It is advantageous for parallelizing the generation of the individual bits of the true random number if several pairs of a first SPAD diode and a second SPAD diode are provided, wherein the emissions signals of the second SPAD diodes are feedable to the processing circuit and the processing circuit outputs a different one of the bits for generating the random number based on the emission signal of each second SPAD diode.

[0044] A pair of a first SPAD diode and a second SPAD diode is generally sufficient for the sequential generation of the individual bits of the true random number, wherein the processing circuit outputs the logical values of the bits for the generation of the random number sequentially.

[0045] A further variant of optical coupling of first and second SPAD diodes can be seen in the fact that single first SPAD diode and several second SPAD diodes are provided

which are optically coupled to the single SPAD diode, wherein the emission signals of the second SPAD diodes are feedable to the processing circuit and the processing circuit outputs a different one of the bits for generating the random number based on the emission signal of each second SPAD diode.

[0046] To be able to detect function errors and to be able to generate a random number despite an error, an error detection unit and a pseudo-random number generator can be provided in an advantageous manner, wherein the processing circuit of the quantum-based generator switches from outputting bits for generating the random number based on the emission signal of the at least one second SPAD diode to outputting a bit generated by the pseudo-random number generator for generating a pseudo-random number, when the error detection unit detects an error in the function of the components responsible for the optical quantum-based process of the quantum process-based generator or an error of the processing circuit thereof. Here, it can be expedient that the error detection unit outputs an error signal representing for example the type and/or the cause of the error when an error is detected. The error detection unit advantageously recognizes not only that an error exists, but also what type of error has occurred. The reaction of the quantum-based generator to such events depends on the application and can mean, for example, that self-tests are carried out. Preferably, however, in the event that the quantum-based generator is faulty, it is switched to a pseudo-random number generator, which generates a pseudo-random number based, for example, on the last bits generated by the quantum-based generator before the error was detected.

[0047] Possible sources of error can be:

[0048] malfunction of supply voltages for individual components of the microcontroller, incorrect signal generation of the first and/or second SPAD diodes,

[0049] malfunction in the optical fiber and/or when coupling the first SPAD diodes and/or the second SPAD diodes to the optical fiber,

[0050] circuit failures in the analog and/or digital part of the microcontroller and the quantum random number generator, respectively,

[0051] incorrect entropy of the supplied random numbers, which can be done by testing, as exemplarily described in David Johnston, Random Number Generators-Principles and Practices, Chapters 8 and 9, 2018, Walter De Gruyter GmbH, Berlin/Boston, ISBN 978-1-5015-1530-2.

[0052] The above-described variant of the microcontroller with error detection unit and pseudo-random number generator can also be realized with microcontrollers that have a quantum-based generator other than the one described above. In this respect, independent protection is claimed for the subject matter of this further development.

[0053] In an advantageous configuration of the disclosure, it may be provided that the memory elements comprise one or more read/write memories RAM and/or one or more writable non-volatile memories, optionally EEPROM memories and/or flash memories and/or one time programmable (OTP) memories, and/or one or more read-only memories and/or one or more non-volatile manufacturer memories, optionally one or more manufacturer ROMs, and/or one or more manufacturer EEPROMs and/or one or more manufacturer flash memories.

[0054] In an expedient configuration of the disclosure, it may be provided that the manufacturer Rom comprises a boot software.

[0055] In a further expedient configuration of the disclosure, it may be provided that a manufacturer firewall is provided between the manufacturer memory and the internal bus.

[0056] In an expedient further development of the disclosure, the microcontroller may be provided with

[0057] a base clock generator,

[0058] a clock generator circuit, and/or

[0059] a reset circuit, and/or

[0060] a power supply or a Vcc circuit with voltage regulators that provide the operating voltages, and/or

[0061] a ground circuit, and/or

[0062] an input/output circuit, and/or

[0063] one or more processing modules,

[0064] wherein the processing modules communicate with the internal bus, and

[0065] wherein the processing modules comprise one or more of the following modules:

[0066] a CRC module (cyclic redundancy check),

[0067] a clock generator module,

[0068] with a DES accelerator and/or an AES accelerator,

[0069] one or more timer modules,

[0070] a safety monitoring and control circuit,

[0071] a data interface, optionally a Universal Asynchronous Receiver Transmitters.

[0072] In a further advantageous configuration of the disclosure, it may be provided that

[0073] the semiconductor substrate comprises a surface,

[0074] the semiconductor substrate comprises a semiconducting material below its surface,

[0075] the surface of the semiconductor substrate comprises a metallization stack,

[0076] the metallization stack comprises a typically structured and optically transparent and electrically insulating layer which is located on the surface of the semiconductor substrate,

[0077] at least a part of this transparent and electrically insulating layer forms the at least one optical fiber,

[0078] the first SPAD diode emits photons from the semiconducting material of the semiconductor substrate into the optical fiber,

[0079] the at least one optical fiber transports such photons to the second SPAD diode.

[0080] In an expedient configuration of the disclosure it may be provided that the optical fiber irradiates the second SPAD diode in such a way that the light from inside the optical fiber re-enters the semiconducting material of the semiconductor substrate and hits device parts of the second SPAD diode there.

[0081] In a further expedient configuration of the disclosure, it may be provided that

[0082] the at least one operating circuit at least temporarily supplies the at least one first SPAD diode with electrical energy,

[0083] the at least one first SPAD diode feeds photons into the at least one optical fiber when supplied with sufficient electrical energy,

[0084] the at least one optical fiber transports such photons to the second SPAD diode, and

[0085] the at least one optical fiber emits such photons into the second SPAD diode.

[0086] In an advantageous configuration of the disclosure, it may be provided that the one data bus interface is a wired automotive data bus interface or that one or more data bus interfaces of the several data bus interfaces are wired automotive data bus interfaces.

[0087] In a further advantageous configuration of the disclosure, it may be provided that the one or more wired automotive data bus interfaces are configured as

- [0088] a CAN data bus interface, and/or
- [0089] a CAN FD data bus interface, and/or
- [0090] a Flexray data bus interface, and/or
- [0091] an PSI5 data bus interface, and/or
- [0092] an DSI3 data bus interface, and/or
- [0093] a LIN data bus interface, and/or
- [0094] an Ethernet data bus interface, and/or
- [0095] a MELIBUS data bus interface.

[0096] In an expedient further development of the disclosure, it may be provided that the one data bus interface is a wireless data bus interface or that one or more data bus interfaces of the data interfaces are wireless data bus interfaces.

[0097] In a further advantageous configuration of the disclosure, it may be provided that the one or more wireless data bus interfaces are configured as

- [0098] a WLAN interface, and/or
- [0099] a mobile radio interface, and/or
- [0100] a Bluetooth interface.

[0101] In an expedient configuration of the disclosure, it may be provided that the one data bus interface is a wired data bus interface or that one or more data bus interfaces of the several data interfaces are wired data bus interfaces.

[0102] In a further expedient configuration of the disclosure, it may be provided that the one or more wired data bus interfaces are configured as

- [0103] a KNX data bus interface, and/or
- [0104] an EIB data bus interface, and/or
- [0105] a DALI data bus interface, and/or
- [0106] a PROFIBUS data bus interface.

[0107] A significant advantage of the disclosure is that the semiconductor material into which the optically coupled SPAD diodes are integrated is not used for transporting the photons, but rather one or more dielectric layers, i.e. insulation layers of the metallization stack, close to the surface of the semiconductor material are used for the transport. This massively increases the “yield” of photons reaching the receiver SPAD diode, which means that more events can be observed in the receiver SPAD diode in a shorter time, which in turn can be utilized to generate longer quantum process-based random numbers in a very short time and a plurality of quantum process-based random numbers in a very short time, respectively. This increases the effectiveness of the generation of quantum process-based random numbers, which are also secure with regard to so-called post-quantum cryptography (PQK) or quantum computer-resistant cryptography.

BRIEF DESCRIPTION OF THE DRAWINGS

[0108] The disclosure is explained in more detail below by means of various examples and with reference to the drawing.

[0109] FIG. 1 shows a block diagram of an example of a secure microcontroller 1.

[0110] FIG. 2 shows an exemplary process for the protection of a product by means of a first secure microcontroller (IC1) in line with the disclosure and by means of a second secure microcontroller (IC2). Here, the first secure microcontroller (IC1) and the second secure microcontroller (IC2) exchange data with each other, which is preferably encrypted using one or more quantum process-based generators for true random numbers.

[0111] FIG. 3 shows the integration of an SPAD diode in a semiconductor material.

[0112] FIGS. 4 and 5 show the combination of a first SPAD diode with a second SPAD diode, both of which are integrated in a semiconductor material and optically coupled by an optical waveguide applied to the semiconductor material as part of the metallization stack, namely as part or section of insulation layers of the metallization stack.

[0113] FIG. 6 shows the combination of a quantum-based entropy source of at least one transmitter SPAD diode and at least one receiver SPAD diode with optical fibers that optically couple them and comprise an evaluation and operating circuit.

[0114] FIG. 7 shows an arrangement similar to that in FIG. 6, but with the addition of a monitoring circuit.

[0115] FIG. 8 shows a flow diagram for an entropy extraction method according to an example of the disclosure.

[0116] FIG. 9 shows a typical output signal of a receiver SPAD diode.

[0117] FIG. 10 schematically shows the individual method steps for generating a quantum random number using the optical quantum-based process according to the disclosure.

DESCRIPTION

[0118] The figures show parts of the proposed devices and methods in an exemplary and simplified manner. For the purpose of illustration, certain examples designed in accordance with the teachings of the disclosure will now be described with reference to the accompanying drawings.

[0119] The examples of the disclosure in the following description and in the drawing are to be regarded as illustrative and are not to be regarded as limiting the specific example or element described. Several examples can be derived from the following description and/or the drawings by modifying, combining or varying certain elements. Furthermore, examples or elements that are not described literally can be derived from the description and drawings by a person skilled in the art.

[0120] An integrated circuit according to the disclosure preferably comprises a secure microcontroller 1, i.e. a microcontroller with secure encryption of data. FIG. 1 shows an example of a secure microcontroller 1 at block diagram level. For example, the secure microcontroller 1 includes memory elements that are connected to an internal bus 2. For example, the memory elements can comprise one or more read/write memories RAM 3 and/or one or more writable non-volatile memories such as EEPROM memories 4 and/or flash memories 4 and/or OTP memories 4. Furthermore, the secure microcontroller 1 preferably comprises one or more non-volatile, read-only memories 5, such as a ROM. In addition, the secure microcontroller 1 preferably comprises one or more non-volatile, writable and/or non-writable manufacturer memories 6, which typically include reference data that may be of importance for potential subsequent examinations of a microcontroller or other electronic device. In the case on a non-writable manufacturer

memory 6, the manufacturer memory 6 can be a manufacturer ROM. Preferably, the manufacturer ROM 6 comprises the boot software. For example, the secure microcontroller 1 comprises one or more cryptographic accelerators 7, for example a DES accelerator and/or an AES accelerator 7, which accelerates cryptographic calculations and is connected to internal bus 2. For example, preferably at least one manufacturer firewall 8 is provided between manufacturer memory 6 and internal bus 2. The microcontroller core 16 accesses said memories via data bus 2. For example, the secure microcontroller 1 comprises processing modules which communicate with microcontroller core 16 via internal bus 2. The processing modules of microcontroller 1 preferably comprise at least one of the following modules: a CRC module (cyclic redundancy check) 11, a clock generator module 12, one or more timer modules 13, a safety monitoring and control circuit 14, one or more quantum process-based generators 15 for true random numbers (Quantum Random Number Generator: QRNG), a 8/16/32/64-bit microcontroller core 16 and one or more data interfaces 17, optionally one or more universal asynchronous receiver transmitter (UART) to support high-speed serial data. The other circuit parts of the secure microcontroller 1 include, for example, one or more base clock generator circuits 21 (CLK) and/or one or more clock generator modules 12, a reset circuit 22, a power supply or Vcc circuit 23 with voltage regulators that provide the operating voltage, a ground circuit 24, and an input/output circuit 25.

[0121] Preferably, the secure microcontroller 1 is configured such that it enables secure authentication. Thus, in addition to an authentication code, the secure microcontroller 1 stores further data, e.g. one or more lifetime and usage duration data and/or e.g. logistical data and/or e.g. commercial data and/or website and email addresses and/or image data, a set of instructions for control units of the motor vehicle with which microcontroller core 16 communicates via a data interface. In addition, the secure microcontroller 1 can store further application data.

[0122] Preferably, the first integrated circuits comprises, for example, a secure microcontroller 1 configured to facilitate secure authentication of a product.

[0123] FIG. 2 shows an example of a method for manufacturing a circuit for a product comprises the secure microcontroller 1 according to the disclosure. For example, the method includes writing a product ID to a second integrated circuit of the product (block 250). The second integrated circuit preferably comprises a second secure microcontroller 1 in line with this application. For example, the method includes writing an authentication code corresponding to the product ID to the memories of the first integrated circuit of the first secure microcontroller 1 (block 211). The first integrated circuit of the product preferably also comprises a secure microcontroller 1 in line with this disclosure. For example, the method includes writing various product IDs and various corresponding authentication codes to the memories of the respective integrated circuit of the respective secure microcontroller 1. Various product IDs and various corresponding authentication codes can also be written to the memories of the respective integrated circuit of the respective secure microcontroller 1 per product (block 221), so that each product has a unique product ID and a unique authentication code. The latter step, for example, provides a secure and unique authentication code for each product.

[0124] For example, some of the features described in this description enable secure authentication of a circuit or product, while at the same time enabling integrated and cost-effective manufacture of that circuit or product. For example, the second integrated circuit is preferably configured such that it enables secure authentication. In various examples, different host devices are able to authenticate the circuit or product, e.g. a car, a smartphone, a web server, any data processing device, etc. In one example, the interface between the host device and the circuit is established via the car's control computer. In another example, a first integrated circuit of the circuit is configured to store additional data such as product-related codes, product setting information, etc. For example, the host device can only access, modify or process such further data once secure authentication has been performed via the first integrated circuit. For example, the first integrated circuit is configured to provide or enable access to the above-mentioned additional data only after authentication. In one example, the product requires little or no adaptation for use in different motor vehicle series.

[0125] The internal data bus 2 (see FIG. 1) can comprises several data buses 2 for several microcontroller cores 16, so that these can access different sub-devices of the secure microcontroller 1 independently of one another with a time delay and/or overlapping or simultaneously. However, the secure microcontroller 1 usually comprises only one internal data bus 2 and only one microcontroller core 16. Preferably, microcontroller core 16 is an advanced risk machine (ARM) processor or the like. Preferably, it is an 8-bit or a 16-bit or a 32-bit or a 64-bit microcontroller computer core.

[0126] Preferably, the secure microcontroller 1 comprises one or more read/write memories RAM 3. This may be SRAMs and/or MRAMs and/or FRAMS or the like. They can also be dynamic read/write memories such as DRAMs, which have to be read and rewritten at regular intervals in a refresh cycle. To access its memory, the secure microcontroller 1 according to the disclosure can have an access logic that regularly executes the refresh process. However, a DRAM typically opens up opportunities for an attack and is typically a potential vulnerability. The microcontroller core 16 can preferably access this read/write memory RAM 3 by means of internal data bus 2.

[0127] Preferably, the secure microcontroller 1 comprises one or more writable and non-volatile memories 4. Microcontroller core 16 can preferably access these writable and non-volatile memories 4 via internal data bus 2. Said non-volatile memories can comprise EEPROM memories 4 or flash memories 4 or OTP memories 4, for example. OTP stands for "one time programmable".

[0128] One attack option can be to erase the non-volatile memories 4 by means of radiation, for example X-rays and/or ionizing radiation and/or heating of memory cells. For this purpose, the secure microcontroller 1 preferably comprises one or more safety monitoring and control circuits 14 which monitor the data integrity of the memory cells of the erasable memories 4. Preferably, the memory cells have redundancy in such a way that at least two check bits are provided for a data word, which is preferably a data word of 8 bits in length, i.e. a byte, and that at least one check bit must always have the content 1 and another the content 0. For example, the first check bit can be a parity bit of the byte and the second check bit can be the inverse bit of the parity bit. If an attack with ionizing radiation or the like now occurs, the attack resets both check bits to the same value.

The one or more safety monitoring and control circuits **14** detect this deviation and block secure microcontroller **1** from further access.

[0129] Preferably, each bit of the memories of the secure microcontroller **1** is designed with redundancy, so that each logical data bit is realized as a pair of a first physical data bit with a first internal logical value and a second physical data bit with a second internal logical value. The second internal logical value is typically the logical inverse of the first internal logical value. The one or more safety monitoring and control circuits **14** preferably monitor that this is always the case. The one or more safety monitoring and control circuits **14** detect deviations and, for example, preferably block the further execution of programs or certain program parts by the microcontroller core **16** and/or block access to data in the event of deviations.

[0130] Preferably, the secure microcontroller **1** comprises one or more reset circuits **22** (see FIG. 1). The reset circuits **22** each reset the secure microcontroller **1** and/or sub-devices of the secure microcontroller **1** to predefined states if predetermined or determinable reset conditions and/or combinations and/or time sequences of such reset conditions are present. For example, said condition can be signaling of one or more safety monitoring and control circuits **14**. Said conditions can also be changes and/or values of the potential of the operating voltage of the secure microcontroller. Furthermore, such conditions may affect the integrity of the housing of the secure microcontroller **1**.

[0131] Preferably, the secure microcontroller **1** comprises a detector for opening the housing of the secure microcontroller **1**. For example, this can be a single line that surrounds or covers the secure microcontroller **1** as a textile network or fabric, or at least covers parts of the secure microcontroller **1**. It can also be a network of lines that cover the secure microcontroller **1** solely for the purpose of detecting an attack. For example, the secure microcontroller **1** can have a first input/output by means of which it can feed an electrical current into such a line and draw it again at a second input/output. If the current flow is interrupted, this is an indication of an attack, which the one or more safety monitoring and control circuits **14** detect and which then signal this attack to microcontroller core **16** of the secure microcontroller **1** according to the disclosure, for example. For example, in such a case of suspected violation of the integrity of the housing, one or more of the one or more safety monitoring and control circuits **14** may disable write and/or read access to memory contents of the memories of the secure microcontroller **1** or erase such contents or set such contents to predefined values or overwrite them with nonsensical data or otherwise manipulate them. The memories of the secure microcontroller **1** preferably comprises one or more non-volatile, read-only memories **5**, such as a ROM. The ROM **6** of the secure microcontroller **1** preferably contains data and/or program instructions defined by the design.

[0132] Preferably, the secure microcontroller **1** (see e.g. FIG. 1) comprises one or more non-volatile, writable and/or non-writable manufacturer memories **6**, in which the semiconductor manufacturer or another supplier can store production and security data, such as serial numbers, etc. Preferably, the semiconductor manufacturer blocks access to this writable and/or non-writable non-volatile manufacturer memory **6** after the last production test has been performed. Preferably, access to the writable and/or non-writable manu-

facturer memory **6** is possible by means of a manufacturer password. In some cases, a double key procedure makes sense. In such a case, a customer (downstream of the semiconductor manufacturer) stores a customer password in a customer blocking register that can also be blocked for access with a password. Preferably, the semiconductor manufacturer can only access all memory areas of the secure microcontroller **1** with the customer password and the manufacturer password. Preferably, the semiconductor manufacturer provides an analysis password by means of which the manufacturer can cause one or more of the one or more safety monitoring and control circuits **14**, typically with the aid of the reset circuit **22**, to erase the customer contents and then make all memory areas of the secure microcontroller **1** accessible for the analysis of errors. In the case of a non-writable manufacturer memory, the manufacturer memory **6** can be a manufacturer ROM, for example, the content of which is determined during the manufacture of the semiconductor circuit of the secure microcontroller, for example.

[0133] The secure microcontroller is typically adapted to receive and/or send encrypted data and/or program code parts and/or instructions by means of cryptographic methods stored in its memories and executed by microcontroller core **16**. Some of these methods require considerable computing power. It has therefore proved useful for microcontroller core **16** not to execute certain program parts of these cryptographic methods in the form of sub-steps of these cryptographic methods, but for one or more special hardware accelerators, preferably in the form of one or more cryptographic accelerators **7**, to execute these program parts at an accelerated rate using specially synthesized hardware logic instead of microcontroller core **16**. For this purpose, the secure microcontroller **1** preferably has a DES accelerator for the Data Encryption Standard (DES) algorithm and/or an AES accelerator **7** for executing the Advanced Encryption Standard (AES) algorithm, for example. The microcontroller core **16** typically addresses these hardware accelerators **7** via internal data bus **2**. Preferably, microcontroller core **16** has a redundant clock system in order to be able to recognize accesses to the clock system. One or more of the one or more security monitoring and control circuits **14** monitor the consistency of the logical contents of this preferred plurality of redundant clock systems, and can thus detect attacks and errors. Preferably, one or more manufacturer memory firewalls **8** prevent microcontroller core **16** and the test logic of the secure microcontroller **1** from accessing the manufacturer memories. They can preferably be unlocked using a manufacturer password as described. Preferably, the number of incorrect entries is very limited in order to minimize the probability of a successful attack.

[0134] Preferably, the secure microcontroller comprises one or more CRC modules (cyclic redundancy check) **11** in order to generate the CRC data for serial data communication, which is used in most data protocols to detect faulty data transmissions, on the one hand in the case of a transmission process and on the other hand to be able to check the correct reception of the data message on receipt. Preferably, the safe microcontroller **1** comprises one or more clock generator modules **12** (clock driver, CLK), which generate one or more clocks for operating the circuits of the secure microcontroller. Preferably, the one or more clock generator modules (clock driver, CLK) generate **12** redundant clocks that indicate an attack on the clock system. Typically, the

secure microcontroller 1 comprises one or more timer modules 13, such as those required for detecting time-outs. Preferably, the secure microcontroller 1 comprises one or more watchdog timers that monitor the execution of the various program parts. Said watchdog timers can be part of the one or more safety monitoring and control circuits 14.

[0135] According to the disclosure (see e.g. FIG. 1), the secure microcontroller 1 comprises at least one quantum process-based generator 15. Quantum-based processes have the advantage that they are based on real chance. In the 1970s, the physicist Bell proved that the theory of “hidden parameters” was wrong. This means that there are no hidden causes for the randomness of quantum mechanical events, such as the emission of photons. For example, microcontroller core 16 can be an 8-bit microcontroller core or a 16-bit microcontroller core or a 32-bit microcontroller core or a 64-bit microcontroller core or a 128-bit microcontroller core or the like. The secure microcontroller 1 can comprise one or more 8/16/32/64/128-bit microcontroller cores 16, which can preferably access the other sub-devices via one or more internal data buses 2. Preferably, the secure microcontroller 1 comprises one or more data interfaces 17. For example, such data interfaces can be one or more universal asynchronous receiver transmitters (UART) to support high-speed serial data. Preferably, the secure microcontroller 1 comprises one or more base clock generators 21 (CLK), each of which preferably provides a base clock 12 to one or more clock generator modules (clock driver, CLK). Preferably, the base clock generators 21 (CLK) are oscillators. Preferably, the secure microcontroller 1 also comprises one or more power supply or Vcc circuits 23 with voltage regulators that provide the operating voltages for the secure microcontroller 1. Preferably, the secure microcontroller 1 also comprises one or more ground circuits 24 (i.e. a circuit inserted in the ground line or a circuit “to ground”, in the simplest case a line) comprising, for example, reverse polarity protection and protection circuits against manipulation of the electrical potential of the semiconductor substrate. For example, it is useful if one or more of the ground circuits 24 have reverse polarity protection. For example, it is useful, if one or more of the ground circuits 24 and/or one or more of the power supply or Vcc circuits 23 interact such that the modulation of the power consumption and/or the internal resistance and/or the voltage drop between the supply voltage terminals of the secure microcontroller does not allow any conclusions to be drawn about the operating processes and/or states of the secure microcontroller, at least temporarily.

[0136] For controlling other devices and/or for communicating with other devices and/or for monitoring other devices, it is generally useful if the secure microcontroller has one or more input/output circuits 25, which are generally designed as digital inputs and/or as digital outputs, which preferably can also assume a tri-state condition. The secure microcontroller 1 can comprise an analog-to-digital converter allowing the secure microcontroller 1 to monitor internal analog values, such as the operating voltage, and external analog values. Possibly, the secure microcontroller 1 can be provided with a driver stage, for example to drive actuators. Said actuators may be motors and/or other resistive and/or inductive and/or capacitive loads or the like. Such a driver stage can be a half bridge and/or an H-bridge

or the like, for example. It is also conceivable that it could be power current sources, for example for lamps such as LEDs.

[0137] Thus, the disclosure (see e.g. FIG. 1) proposes a secure microcontroller 1 for controlling devices in automobiles, which comprises a semiconductor substrate. Preferably, the secure microcontroller 1 is manufactured using a CMOS circuit technology or a bipolar circuit technology or a BiCMOS circuit technology. The secure microcontroller preferably comprises memory elements, one or more internal data buses 2, one or more 8/16/32/64-bit microcontroller cores 16, one or more data interfaces, and one or more quantum process-based generators 15. Said one or more quantum process-based generators 15 distinguish the secure microcontroller 1 proposed herein from prior art, which is based on so-called true random number generators, which show a poorer von Neumann entropy of the random numbers generated, which are accordingly “less random”.

[0138] Internal data bus 2 can comprise several data buses. The memory elements of the secure microcontroller 1 are typically connected to internal data bus 2. The data interfaces are typically also connected to internal data bus 2. The one or more quantum process-based generators 15 are preferably also connected to internal data bus 2. The one or more microcontroller cores 16 are preferably also connected to internal data bus 2. The one or more quantum process-based generators 15 preferably and typically generate one or more random numbers at request of microcontroller core 16. In contrast to the random numbers of the true random number generators from prior art, these are characterized by a particularly advantageous entropy. Preferably, one or more microcontroller cores 16 generate one or more keys using a respective program from one or more of its memory elements and using one or more of the generated random numbers. Typically, the one or more microcontroller cores 16 encrypt and/or decrypt data, which said microcontroller cores 16 typically exchange with devices external to the secure microcontroller via one or more data interfaces, using a respective program of the respective microcontroller core 16, each originating from one or more of its memory elements, using a respective key of the possibly multiple keys. Typically, the semiconductor substrate substantially integrally comprises all sub-devices of the secure microcontroller 1.

[0139] In a first further development of the secure microcontroller 1 (see FIG. 1), the memory elements of the secure microcontroller 1 comprise one or more read/write memories RAM 3 and/or one or more writable non-volatile memories 4, for example EEPROM memories 4, and/or flash memories 4 and/or OTP memories 4 and/or one or more read-only memories and/or one or more non-volatile manufacturer memories. For example, the one or more manufacturer memories can comprise one or more manufacturer ROMs 6 and/or one or more manufacturer EEPROMs and/or one or more manufacturer flash memories.

[0140] In a second further development, the manufacturer memory, for example a manufacturer ROM 6, comprises the boot software to safely start the secure microcontroller.

[0141] In a third further development (see FIG. 1), the secure microcontroller 1 comprises a manufacturer memory firewall 8 between the manufacturer memory 6 and the internal bus 2, which prevents access to the manufacturer memory without authentication.

[0142] In a fourth further development (see FIG. 1), the secure microcontroller 1 comprises one or more of the following components: a base block generator 21 (CLK), a clock generator circuit 12, a reset circuit 22, a power supply or a Vcc circuit 23 with voltage regulators providing operating voltages, a ground circuit 24, an input/output circuit 25, and one or more processing modules. The processing modules communicate with internal data bus 2 and thus typically with a microcontroller core 16. The processing modules preferably comprise one or more of the following components: a CRC module (cyclic redundancy check) 11, a clock generator module 12, a crypto accelerator, for example a DES accelerator and/or an AES accelerator 7, one or more timer modules 13, one or more security monitoring and control circuits 14, one or more data interfaces, for example one or more universal asynchronous receiver transmitters (UART) 17 (see FIG. 1).

[0143] In a fourth further development of the secure microcontroller 1 (see FIGS. 4 and 5), the latter comprises at least one first SPAD diode 44 and at least one second SPAD diode 45, at least one optical fiber 50, at least one processing circuit, and at least one operating circuit. The circuit integration of a SPAD diode (e.g. the SPAD diode 44 or 45) is shown again in FIG. 3. In said fourth further development according to FIG. 4, the quantum process-based generator 15 (see FIG. 1) comprises at least the first SPAD diode 44 as a source for the optical quantum signal and the second SPAD diode 45 as a photodetector for the optical quantum signal. Furthermore, in said fourth further development, the quantum process-based generator 15 comprises at least the processing circuit and the optical fiber 50. In said fourth further development, the at least one optical fiber 50 optically couples the at least one first SPAD diode 44 with the at least one second SPAD diode 45. The operating voltage supplies the first SPAD diode 44 with electrical energy in such a way that the first SPAD diode 44 emits photons. This is the case if the first SPAD diode 44 has a sufficient electrical bias voltage. In said fourth further development, the processing circuit detects the (output) signal of the second SPAD diode 45 and forms the random number therefrom.

[0144] Whereas in the exemplary example of FIG. 4 it is assumed that the optical fiber 50 is formed by the one electrical insulation layer applied to the surface 46 of the semiconductor substrate, FIG. 5 shows an exemplary example in which, for example, two optically transparent and electrically insulating insulation layers 34', 34" form the optical fiber 50.

[0145] The processing circuit then preferably provides the random number thus formed to one or more of the one or more microcontroller cores 16 via internal data bus 2 (FIG. 1). The configuration of said processing circuit will be discussed below in conjunction with FIGS. 6 to 9.

[0146] In a sixth further development of the disclosure (see FIGS. 4 and 5), the semiconductor substrate comprises a surface 46. Typically, the semiconductor substrate comprises a semiconducting material below its surface 46. In particular when using conventional manufacturing processes for semiconductor circuits, such as CMOS processes, bipolar processes and BiCMOS processes, there is typically a metallization stack of structured metal layers on surface 46 of the semiconductor substrate, and said metal layers have electrically insulating insulation layers thereon, wherein at least one of said insulation layers 34 forms the optical fiber

50 at least in part or in sections. The structured metal layers typically form conductive paths that are electrically separated from each other by the insulating layers. Thus, the metallization stack comprises a typically structured and optically transparent and electrically insulating layer 34 of silicon oxide, for example, which is located on the surface 46 of the semiconductor substrate. At least a part of the insulating layer 34 (i.e. a participant of the insulation layer in lateral and/or vertical extension) on the surface 46 of the semiconductor substrate preferably forms the optical fiber 50. The part of the metallization stack located above this insulation layer 34 is shown schematically in FIGS. 4 and 5. The first SPAD diode 44 typically emits light 47 from the semiconducting material of the semiconductor substrate into said optical fiber 50. This means that, in contrast to prior art, the first SPAD diode 44 generally radiates upwards perpendicular to surface 46 and not laterally into the semiconductor substrate, which has a high attenuation. This allows the device to couple more photons of the first SPAD diode 44 directly to the second SPAD diode 45. The optical fiber 50 transports said photons 48 of the first SPAD diode 44 in the optical fiber 50 to the second SPAD diode 45 with virtually no loss compared to prior art. The lowest level of the metallization stack 43 is used to reflect light (photons). The optical fiber 50 irradiates the second SPAD diode 45 with the photons 48 of the first SPAD diode 44 in such a way that the light 49 from inside the optical fiber 50 re-enters the semiconducting material of the semiconductor substrate from the surface 46 and hits device parts of the second SPAD diode 44 there. The second SPAD diode 45 then generates an output signal as a function of the irradiation with the photons 48 (of light 49), from which the quantum process-based random number is generated, which is described further below.

[0147] Typically, at least one operating circuit at least temporarily supplies the at least one first SPAD diode 44 with electrical energy. The at least one first SPAD diode 44 then feeds photons 47 into the at least one optical fiber 50 when supplied with sufficient electrical energy. The optical fiber 50 then transports said photons 48 further. The at least one optical fiber 50 then radiates the photons 48 as photons 49 substantially "from above" into the second SPAD diode 45. Since this transport of photons from the first SPAD diode 44 to the second SPAD diode 45 loses significantly fewer photons due to the low attenuation in the optical fiber 50 than in the construction of prior art, which uses the strongly absorbing semiconductor substrate, the quantum efficiency is massively higher. Therefore, in the construction presented herein, a pair of a single first SPAD diode 44 and a single second SPAD diode 45 is sufficient (although, nevertheless, several pairs of SPAD diodes can also be used according to the disclosure). The prior art always uses several SPAD diodes on both the transmitter and receiver side.

[0148] FIG. 5 shows that two or more dielectric layers, i.e. optically transparent insulating layers 34', 34", which at least in some areas directly adjoin one another, i.e. are not separated by metallization layers in these areas, form the optical fiber 50'. The contacts 51 and 52 of the SPAD diodes 44 and 45 are positioned in this example in such a way that they are arranged in pairs on both sides of a center region of the optical fiber 50', so that the optical fiber transports photons 47 of SPAD diode 44 in this section of the insulation layers 34', 34" adjacent to each other at a plane 53, as indicated at 48, not necessarily only in the upper of the two

insulation layers 34', 34", but also by means of both, until they meet SPAD diode 45 at 49. The contacts 51, 52 advantageously also serve to reflect and thus "introduce" the photons into SPAD diode 45. In this respect, the contacts 51, 52 of SPAD diode 44 also serve to reflect and "redirect" the photons 47 emerging from SPAD diode 44 in the direction of SPAD diode 45.

[0149] In another further development of the secure microcontroller 1 according to the disclosure, at least one data interface of the one or more data interfaces is a wired automotive data bus interface. In this case, the wired automotive data bus interface can be, for example, a CAN data bus interface or a CAN FD data bus interface or a Flexray data bus interface or a PSI5 data bus interface or a DS13 data bus interface or a LIN data bus interface or an Ethernet data bus interface or a or a MELIBUS data bus interface.

[0150] In another further development of the secure microcontroller 1 according to the disclosure, at least one data interface is a wireless data bus interface. The wireless data bus interface can be a WLAN interface or a Bluetooth interface, for example.

[0151] In another further development of the secure microcontroller 1 according to the disclosure, at least one data interface is a wired data bus interface. The wireless data bus interface can be a KNX data bus interface or an EIB data bus interface or a DALI data bus interface or a PROFUBUS data bus interface, for example.

[0152] FIG. 6 schematically shows the simplified block diagram of a quantum-based random number generator QRNG 400 according to an example of the disclosure. The digital circuits of the exemplary device shown in FIG. 6 are preferably clocked with a preferably common clock. The structure includes a quantum process-based entropy source 401, a preferably broadband radio frequency amplifier 402, an analog-to-digital converter 403 having a resolution of e.g. 14 bits and a sampling rate of e.g. 125 MS/s, and a field programmable gate array (FPGA) 404.

[0153] The entropy source 401 comprises a 2D array of single photon avalanche diodes (SPAD) 401.1 and 401.3. Said SPAD diodes are operated in Geiger mode with a supply voltage above the breakdown voltage. In addition, a quenching resistor 401.4 is connected in series with each SPAD diode. The quenching resistor 401.4 prevents a thermal destruction of the diode in the event of a triggered charge carrier avalanche. The current signal of the SPAD diodes is measured via a shunt resistor, which can be the quenching resistor or a resistor provided in addition thereto. In the example of FIG. 6, the array of SPAD diodes consists of, for example, four (active or transmitter) SPAD diodes 401.1 and twelve (passive or receiver) SPAD diodes 401.3, which are coupled to the SPAD diodes 401.1 via an optical waveguide 401.2, as described above with reference to FIGS. 4 and 5 for the optical fiber 50. The active SPAD diodes 401.1 emit light. They correspond to the first SPAD diode 44 of FIGS. 4 and 5. The active SPAD diodes 401.1 are preferably located inside the array of SPAD diodes. The device as proposed supplies the active SPAD diodes 401.1 with an increased supply voltage and therefore operates the active SPAD diodes 401.1 well above the breakdown voltage. This increases the dark count rate, which leads to a higher number of spontaneously emitted photons 47. The optical waveguide 401.2 forwards some of said photons as photons 48 to the passive SPAD diodes 401.3. For example, the optical waveguide 401.2 corresponds to the optical fiber

50 of FIGS. 4 and 5. Each of the passive SPAD diode 401.3 corresponds to the second SPAD diode 45 of FIGS. 4 and 5, for example. The device as proposed supplies the passive SPAD diodes 401.3 with an increased supply voltage and operates the passive SPAD diodes 401.3 slightly above the breakdown voltage. Preferably, the passive SPAD diodes 401.3 are arranged as a ring around the active SPAD diode 401.2. The passive SPAD diodes 401.3 detect the photons arriving via waveguide 401.2. Depending on the arriving photons, the passive SPAD diodes 401.3 generate a current flow through a shunt resistor, for example, and thus a voltage signal.

[0154] The voltage signal 405 of the entropy source 401 preferably fed to a broadband, e.g. 40 dB high-frequency amplifier 402. The high-frequency amplifier 402 preferably has a bandwidth of 30 to 4000 MHz and preferably a 1 dB compression point of 20 dBm. The voltage swing of voltage signal 405 of entropy source 401 is typically in the sub-millivolt range. The high-frequency amplifier 402 amplifies the voltage swing of the voltage signal 405 of entropy source 401 to e.g. 50 to 150 mV.

[0155] The amplifier output signal 406 of high-frequency amplifier 402 is passed on to an FPGA 404, for example, after an analog-to-digital conversion in ADC 403. Other discrete or ASIC-based solutions are of course possible. In this respect, the FPGA is only one of many different realizations of the technical teaching presented in this example. The FPGA 404 preferably comprises a microcontroller. For example, FPGA 404 can be a Zynq 7010 from Xilinx with a dual-core ARM Cortex-A9 MPCore. This is part of the SPAD evaluation circuit of this exemplary example, which furthermore in the examples of FIGS. 4 and 5 has ADC 403 having, for example, 14 bits with an exemplary sampling rate of 125 mega-samples/s and an exemplary bandwidth of 50 MHz. The amplified voltage signal is available at the input of the high-frequency amplifier 402 as amplifier output signal 406. ADC 403 samples the amplifier output signal 406 of high-frequency amplifier 402. For example, ADC 403 transmits the determined samples of amplifier output signal 406 of high-frequency amplifier 402 digitally with a bit width of e.g. 14 bits to the built-in FPGA 404 of the measurement board.

[0156] The device shown as a block diagram in FIG. 6 in a simplified manner includes a comparator 404.2, a time-to-digital converter (TDC) 404.3, an entropy extraction device 404.4, and a finite-state machine 404.8.

[0157] For example, comparator 404.2 compares the digital 14-bit value 407 of ADC 403 with a reference value 404.1, which represents a threshold, resulting in a two-clock 1-bit output pulse as output signal 409 of comparator 404.2 if the value at the output of ADC 403 is greater than reference value 404.1. The output signal 409 of the comparator 404.2 is fed to the time-to-digital converter 404.3. The time-to-digital converter 404.3 preferably has a 32-bit counter, for example, that counts up in time with the SPAD evaluation circuit. The bit width of the counter may vary depending on the application. This clock can have a frequency of 125 MHz, for example. The 1-bit output signal of comparator 404.2 preferably resets the counter reading of this counter. The time-to-digital converter 404.3 transmits the counter reading present at this point in time, i.e. the counter reading present immediately before the reset, to its output 410. The count result has a resolution of $1/125 \text{ MHz} = 8 \text{ ns}$ for an exemplary 125 MHz clock. The output 410 of the

time-to-digital converter (TDC) **404.3** passes the exemplary 32-bit count result, also referred to as raw data RD, of time-to-digital converter **404.3** to entropy extraction device **404.4**. The entropy extraction device **404.4** converts the raw data RD of TDC **404.3** of the signal of output **410** of time-to-digital converter (TDC) **404.3**, which is random with respect to its composition and sequence, into a 1-bit random number RN **411** at the output of entropy extraction device **404.4**. The output **411** of entropy extraction device **404.4** is connected to the input of a finite-state machine (FSM) **404.8**.

[0158] FSM **404.8** has the task of receiving data from entropy extraction device **404.4** and generating the QRNG random number therefrom, which FSM **404.8** stores in a memory RAM **404.9** of FPGA **404**. After a successful writing operation, FSM **404.8** sets a finish flag **404.10**. The finish flag **404.10** is not set at system startup. A microcontroller **404.11**, for example a dual-core Arm Cortex-A9 MPCore, accesses the RAM **404.9** block and reads the random number from RAM **404.9**. The microcontroller **404.11** is thus e.g. the microcontroller core according to the disclosure.

[0159] FIG. 7 shows the extended exemplary FPGA design with monitoring of the signal to be used for the random number RN at output **411** of entropy extraction device **404.4** and an additional backup system in case a potential error has occurred during the generation of the QRNG-based random number.

[0160] The extension of the device according to FIG. 7 relates to an additional watchdog **404.5**, a linear feedback shift register **404.6** as an example of a PRN generator, a signal multiplexer **404.7** and a voltage monitor **413**. The **404.11** microcontroller can also be set up externally with these components.

[0161] The output **411** of entropy extraction device **404.4** is now connected to watchdog **404.5** and signal multiplexer **404.7**. The watchdog **404.5** monitors the 1-bit random number RN at output **411** of entropy extraction device **404.4** for validity. The watchdog **404.5** detects at least three defined error cases. For example, watchdog **404.5** sends the last valid random numbers as a seed S output signal **412** to the linear feedback shift register **404.6**. If an error occurs, the watchdog sets error bits in an unsigned error register ER of microcontroller **404.11**. Which error bit the watchdog **404.5** sets in the error register of microcontroller **404.11** is preferably dependent on the type of error. In addition, watchdog **404.5** is connected to a voltage monitor **413** via one or more, preferably digital, input-output signal lines **414**.

[0162] For example, voltage monitor **413** monitors the operating voltages of entropy source **401**. If the operating voltage of one of the circuits of the SPAD diodes **401.1** and **401.3** is too low, i.e. the amount is below a lower SPAD operating voltage threshold, or too high, i.e. the amount is above an upper SPAD operating voltage threshold, voltage monitor **413** detects this voltage deviation. In the event of such a voltage deviation, voltage monitor **413** signals this to watchdog **404.5** or directly to microcontroller **404.11**. In the case of signaling to watchdog **404.5**, for example, watchdog **404.5** can generate an interrupt signal for microcontroller **404.11**. For example, watchdog **404.5** can trigger such an interrupt of microcontroller **404.11** or another sub-device of an application system if the supply voltage of entropy source

401 or radio frequency amplifier **402** or another device part of the quantum random number generator QRNG **400** is faulty.

[0163] If watchdog **404.5** has detected an error, quantum random number generator **400** switches to an emergency state. For this purpose, watchdog **404.5** sets selection signal **416** of signal multiplexer **404.7** so that signal multiplexer **404.7** instead of output **411** of entropy extraction device **404.4** applies a pseudo-random number PRN generated by the linear feedback shift register **404.6** in the form of a stream of pseudo-random bits via a pseudo-random signal line **417** as a substitute for the potentially erroneous 1-bit random number RN present at output **411** of entropy extraction device **404.4** to input of FSM **404.8**.

[0164] The linear feedback shift register **404.6** is connected to output **412** of watchdog **404.5** and receives its seed S output signal **412** therefrom. In the event of an error, watchdog **404.5** activates the linear feedback shift register **404.6** (see connection **418**). The linear feedback shift register **404.6** then generates pseudo-random numbers PRN. The seed S of watchdog output signal **412** preferably comprises the last, for example, 16 random numbers that are still valid (e.g. respectively 1 bit). The watchdog **404.5** preferably applies said last valid random numbers to the input of linear feedback shift register **404.6**. The seed S thus serves as a random PQC safe start value for the generator polynomial of the feedback of linear feedback shift register **404.6** for the generation of the pseudo-random number PRN for the pseudo-random signal line **417**. The generator polynomial and the degree of the generator polynomial are preferably freely selectable.

[0165] The signal of output **411** of entropy extraction device **404.4** with the 1-bit random number RN of entropy extraction device **404.4** and the signal of the pseudo-random signal line **417** with the pseudo-random number PRN of linear feedback shift register **404.6**, respectively, are connected to the inputs of signal multiplexer **404.7**. The signal multiplexer **404.7** forwards one of its two input signals to FSM **404.8** depending on the value SEL of selection signal **416**. It is conceivable to use a multiplexer with more than two inputs and a more complex control signal if the application requires this. The number of inputs of signal multiplexer **404.7** is therefore typically greater than or equal to two.

[0166] FSM **404.8** in turn has the task of receiving the random data RN or the pseudo-random number PRN from the output of signal multiplexer **404.7** and writing them to the memory RAM **404.9** of FPGA **404**. If the writing process is successful, FSM **404.8** sets the finish flag **404.10** again. The microcontroller **404.11** can then access the memory **404.9** RAM and read the random number and use it for encryption, authentication, signing, etc., for example.

[0167] FIG. 8 shows a flow diagram **500** for an entropy extraction method which is carried out, for example, by entropy extraction device **404.4** of QRNG **400** of FIGS. 6 and 7. The method provides for first determining, in a first step **501**, two values of output **410** of time-to-digital converter **404.3** and storing them in a shift register of entropy extraction device **404.4**. If two values are stored in the shift register of entropy extraction device **404.4**, entropy extraction device **404.4** compares these two values in a second step **502**. The two values in the shift register of entropy extraction device **404.4** thus comprise a first value and a second value, both of which were determined by time-to-digital converter

404 by means of two different measurements of the respective period between two signal pulses above reference value **404.1**. In a third step **503**, entropy extraction device **404.4** evaluates the two values. If the first value is smaller than the second value and the difference between value **1** and value **2** is greater than a minimum difference **E**, entropy extraction device **404.4** sets the value of its output **411** to a first logical value. If the first value is greater than the second value and the difference between the first value and the second value is greater than the minimum difference **E**, entropy extraction device **404.4** sets its output to a second logical value that is different from the first logical value.

[0168] If the difference between the first value and the second value is smaller than the minimum difference **E**, the entropy extraction discards the first value and the second value. In such a case, the entropy extraction method preferably causes the watchdog (in the device of FIG. 7) to increment an error counter by a first error counter increment. The first error counter increment can be negative. Conversely, entropy extraction device **404.4** can decrease the error counter of the watchdog by a second error counter increment if the difference between the first value and the second value is greater than the minimum difference **E**. The second error counter increment can be the same as the first error counter increment. Typically, the signs of the first error counter increment and the second error counter increment are the same. Preferably, microcontroller **404.11** can set the error counter increments and the start value of the error counter and an error counter threshold value. If the count reading of the error counter crosses the error counter threshold, watchdog **404.5** signals the presence of a critical error state, preferably by means of an interrupt or another signaling to the microcontroller. The microcontroller **404.11** then typically starts a self-test program to test the various parts of the quantum random number generator **400** according to FIG. 7. Preferably, the microcontroller can, for example, set the analog-to-digital converter **403** to a state in which microcontroller **404.11** can write test values to an output register of the analog-to-digital converter, which the subsequent signal chain then processes like real sampled values. Since the test values are known in advance, the correct reaction of the rest of the system, for example the incrementing of the error counter in watchdog **404.5** by microcontroller **404.11**, can be monitored and evaluated. Preferably, microcontroller **404.11** can therefore monitor all memory nodes of FPGA **404** and read their logical state.

[0169] If a value is less than a minimum value, it is a value that is within the dead time of the SPAD diodes. Such a value is preferably discarded, and the error counter is incremented by the first error increment. In this case, entropy extraction device **404.4** waits for the time-to-digital converter **404.3** to determine the next value.

[0170] Once the random bit has been extracted in this way, the method starts again from the beginning.

[0171] If the error counter crosses or reaches the error counter threshold, there may be an error in which the time-to-digital converter supplies constant numbers, for example.

[0172] This device is therefore able to detect a failure of the power supply of the entropy source or other parts of the device. The microcontroller **404.11** can also record ADC **403** for test purposes and voltages and currents in the quantum random number generator **400** and compare the values determined in this way with expected value ranges

within which these values must lie. The microcontroller **404.11** can also record digital values within quantum random number generator **400**. For example, microcontroller **404.11** can set reference value **404.1** so low for test purposes that time-to-digital converter **404.3** is substantially controlled by the signal noise at its input. The values of time-to-digital converter **404.3** should then satisfy the expected statistics within a tolerance band. If this is not the case, there is an error.

[0173] The watchdog **404.5** can monitor the entropy of the random numbers provided. If the average entropy of the bits over an entropy measurement period deviates by more than a permitted entropy deviation value of e.g. 50%, the watchdog concludes that an error has occurred and increments the error counter. Preferably, the watchdog then stops the use of these random bits at output **411** of entropy extraction device **404.4** to prevent the sending of plaintext. "Plain text" refers to information that can be understood by a third party using statistical methods or directly, as the information is only poorly encrypted. It is conceivable that a logical "permanent one" or a logical "permanent zero" is generated by chance even with functioning sub-devices. It therefore makes sense if the maximum length of a bit sequence at the output of the entropy extraction device is limited to a value that can be programmed by microcontroller **404.11**.

[0174] Substantially, the quantum random number generator **400** described above can thus recognize the following errors and catch them with a lower safety level by means of an emergency mode using a linear feedback shift register **404.6** or another PRN generator:

[0175] malfunction of supply voltages,

[0176] incorrect signal generation of the SPAD diodes **401.1** and **401.3**,

[0177] malfunction of optic fiber **401.2** and/or the couplings of the SPAD diodes **401.1** and **401.3** to optical fiber **401.2**,

[0178] circuit failures in the FPGA **404**, i.e. in the digital part of the quantum random number generator **400**,

[0179] incorrect entropy of the supplied random numbers, which can be done by testing the 1-bit random number RN, as described in e.g. David Johnston, Random Number Generators-Principles and Practices, Chapters 8 and 9, 2018, Walter De Gruyter GmbH, Berlin/Boston, ISBN 978-1-5015-1530-2.

[0180] It is conceivable to use a second complete quantum random number generator **400** instead of the linear feedback shift register **404.6**, whose signal at the output of its entropy extraction device **404.4** is then used for emergency mode instead of the signal of the pseudo-random signal line **417**.

[0181] FIG. 9 shows an exemplary oscillogram of voltage signal **405** at the output of entropy source **401**. As can be seen, first spikes occur with a first height class **601** and second spikes with a second height class **602**. The scatter of the first height class **601** of the first spikes and the scatter of the second height class of the second spikes is so small in each case that a clear separation of both height classes **601**, **602** is possible by means of a cutting level **603**. The cutting level **603** corresponds to the value that microcontroller **404.11** sets as reference value **404.1**. For the formation of the first and second spikes as a result of the simultaneous occurrence of a spontaneous and a stimulated photon emission of the second SPAD diode or one of the second SPAD diodes (second spikes) and as a result of exclusively to a

spontaneous photon emission of the second SPAD diode or one of the second SPAD diodes (first spikes), please refer to the explanation above in the test.

[0182] FIG. 10 schematically shows the method 3700 as proposed for generating a quantum random number. The method 3700 starts with generating 3710 a random single photon current (47, 48, 49, 401.2) by means of one or more first SPAD diodes (401.1, 44). The method 3700 continues with transmitting 3720 the random single photon current (47, 48, 49, 401.2) by means of an optical fiber (44, 401.2) different from the semiconductor substrate (49, 48) to one or more second SPAD diodes (401.3, 45). This is followed by converting 3730 the random single photon current (47, 48, 49, 401.2) into a detection signal in the form of a voltage signal 405 of the entropy source 401, which preferably comprises the first SPAD diodes 401.1 and the optical fiber 401.2 and the second SPAD diodes 401.3. This is then followed by conditioning 3740, optionally amplifying and/or filtering and/or analog-to-digital converting, the detection signal into a conditioned detection signal, optionally a digital 14-bit value 407 of the analog-to-digital converter 403. Then separating 3750 the pulses of the conditioned detection signal produced by coupling the emissions of a first SPAD diode 401.1 and a second SPAD diode 401.3 from the pulses of the conditioned detection signal produced exclusively by spontaneous emission by comparing the conditioned detection signal with a threshold value, optionally in a comparator (see 404.2 in FIGS. 6 and 7), and generating a corresponding output signal 409, optionally of the comparator 404.2. This is followed by determining 3760 a first time interval between the first pulse and the second pulse of a first pair of two successive photons produced by couplings of the spontaneous photon emission of a first SPAD diode 401.1 with a photon emission of the second SPAD diode 401.3 stimulated by the photon emission, and determining a second time interval between a third pulse and a fourth pulse of a second pair of two successive pulses of the conditioned detection signal produced by the same kind of couplings of the emissions of a first SPAD diode 401.1 and a second SPAD diode 401.3 and, optionally, for determining the first value of the output 410 of the time-to-digital converter 404.3 and the second value of the output 410 of the time-to-digital converter 404.3. The bit value of a random bit is then determined 3670 on this basis by comparing the value of the first time interval and the value of the second time interval. The last step 3680 checks whether the number n of random bits determined is still smaller than the number m of random bits of the desired quantum random number. If this is not the case, the above steps are repeated. Otherwise, the process of generating a quantum random number is complete.

[0183] The secure microcontroller presented herein has an improved entropy of its at least one random number generator. As a result, the encryption that can be achieved with the microcontroller is more effective compared to prior art and is also post quantum secure. However, the advantages of the disclosure are not limited thereto.

[0184] The above description does not claim to be exhaustive and is not limited to the examples shown. Other variations on the examples described herein may be understood and practiced by those of ordinary skill in the art with reference to the drawings, description and claims. The indefinite singular article "a" does not exclude the plural, while mentioning a certain number of elements does not

exclude the possibility that there are more or fewer elements. A single unit can fulfill the functions of several elements mentioned in the description, and vice versa, several elements can fulfill the function of one unit. Numerous alternatives, equivalents, variations and combinations are possible without departing from the scope of the present application.

[0185] Unless otherwise indicated, all features of the present disclosure can be freely combined with one another. This applies to the entire application presented herein. Unless otherwise indicated, the features described in the figure description can also be freely combined with the other features as features of the disclosure. A restriction of individual features of the examples to the combination with other features of the examples is expressly not intended. In addition, device features of the device can also be reformulated as method features and method features can be reformulated as device features of the device. Such a reformulation is thus also automatically disclosed.

[0186] In the preceding detailed description, reference is made to the attached drawings. The examples in the description and the drawings are to be regarded as illustrative and are not to be regarded as limiting the specific example or element described. Several examples can be derived from the proceeding description and/or the drawings and/or the claims by modifying, combining or varying certain elements. Furthermore, examples or elements that are not described literally can be derived from the description and/or the drawings by a person skilled in the art.

[0187] The disclosure was described above with reference to a microcontroller for automotive applications. However, it is clear to the person skilled in the art that the intended use in this respect based on the disclosure both in the patent application(s) giving rise to a right of priority and in the present PCT application is not restrictive. Rather, the microcontroller according to the disclosure can be used in all devices involving PQR-secure encryption and/or the PQR signature of data and the editing and processing of security-relevant data. A PQR encryption can preferably be carried out using one of the following methods:

[0188] BIKE1-L1-CPA, BIKE1-L3-CPA, BIKE1-L1-FO, BIKE1-L3-FO, Kyber512, Kyber768, Kyber1024, Kyber512-90s, Kyber768-90s, Kyber1024-90s, LEDAcryptKEM-LT12, LEDAcryptKEM-LT32, LEDAcryptKEM-LT52, NewHope-512-CCA, NewHope-1024-CCA, NTRU-HPS-2048-509, NTRU-HPS-2048-677, NTRU-HPS-4096-821, NTRU-HRSS-701, LightSaber-KEM, Saber-KEM, FireSaber-KEM, BabyBear, BabyBearEphem, Mama-Bear, Mama-BearEphem, PapaBear, PapaBearEphem, FrodoKEM-640-AES, FrodoKEM-640-SHAKE, FrodoKEM-976-AES, FrodoKEM-976-SHAKE, FrodoKEM-1344-AES, FrodoKEM-1344-SHAKE, SIDH-p434, SIDH-p503, SIDH-p610, SIDH-p751, SIDH-p434-compressed, SIDH-p503-compressed, SIDH-p610-compressed, SIDH-p751-compressed, SIKE-p434, SIKE-p503, SIKE-p610, SIKE-p751, SIKE-p434-compressed, SIKE-p503-compressed, SIKE-p610-compressed, SIKE-p751-compressed.

[0189] A PQR signature can preferably be created using one of the following methods:

[0190] DILITHIUM_2, DILITHIUM_3, DILITHIUM_4, MQDSS-31-48, MQDSS-31-64, SPHINCS+-Haraka-128f-robust, SPHINCS+-Haraka-128f-simple,

SPHINCS+-Haraka-128s-robust, Sphincs+-Haraka-128s-simple, Sphincs+-Haraka-192f-robust, Sphincs+-Haraka-192s-robust, Sphincs+-Haraka-192s-simple, Sphincs+-Haraka-256f-robust, Sphincs+-Haraka-256f-simple, Sphincs+-Haraka-256s-robust, Sphincs+-Haraka-256s-simple, Sphincs+-SHA256-128f-robust, Sphincs+-SHA256-128f-simple, Sphincs+-SHA256-128s-robust, Sphincs+-SHA256-128s-simple, Sphincs+-SHA256-192f-robust, Sphincs+-SHA256-192f-simple, Sphincs+-SHA256-192s-robust, Sphincs+-SHA256-192s-simple, Sphincs+-SHA256-256f-robust, Sphincs+-SHA256-256f-simple, Sphincs+-SHA256-256s-robust, Sphincs+-SHA256-256s-simple, Sphincs+-SHAKE256-128f-robust, Sphincs+-SHAKE256-128f-simple, Sphincs+-SHAKE256-128s-robust, Sphincs+-SHAKE256-128s-simple, Sphincs+-SHAKE256-192f-robust, Sphincs+-SHAKE256-192f-simple, Sphincs+-SHAKE256-192s-robust, Sphincs+-SHAKE256-192s-simple, Sphincs+-SHAKE256-256f-robust, Sphincs+-SHAKE256-256f-simple, Sphincs+-SHAKE256-256s-robust, Sphincs+-SHAKE256-256s-simple, picnic_L1_FS, picnic_L1_UR, picnic_L3_FS, picnic_L3_UR, picnic_L5_FS, picnic_L5_UR, 'picnic2_L1_FS', picnic2_L3_FS, picnic2_L5_FS, qTesla-p-I, qTesla-p-III'.

[0191] For example, the concept according to the disclosure can be used to generate the addresses in a communication bus system with automatic addressing of the subscribers by the bus master in the bus master or in the subscribers to be addressed. Thus, the disclosure can be used not only for encryption techniques for generating PQK-secure codes. PQK stands for "post-quantum cryptography" and refers to a subfield of cryptography, namely the subfield of quantum computer-resistant cryptography, which deals with cryptographic primitives that, unlike most currently used asymmetric cryptosystems, are practically impossible to decrypt even using quantum computers (see the definition in Wikipedia). Different codes can also be used to better distinguish the signals emitted by a system of basically any design in order to separate these signals and to separate these signals from neighboring systems that may interfere with each other in terms of signal technology. If you think of an ultrasonic measuring device in the automotive sector, for example, it can be advantageous to make the individual received signals distinguishable with regard to their origin, i.e. with regard to the emitting transmitter from which the signals originate.

[0192] The following listing of the disclosure summarizes the features of the disclosure and its further developments once again. Applications of the technical teaching can combine the features with each other, provided that these combinations do not cause factual contradictions. In this respect, the dependencies and references presented herein only represent particularly preferred, exemplary examples.

[0193] Item 1) A secure microcontroller 1 for controlling devices in an automobile

[0194] comprising a semiconductor substrate, and

[0195] comprising memory elements, and

[0196] comprising at least one internal bus 2, and

[0197] comprising at least one 8/16/32/64-bit microcontroller core 16, and

[0198] comprising one or more data interfaces, and

[0199] comprising at least one quantum process-based generator 15, and

[0200] wherein the memory elements are connected to internal bus 2, and

[0201] wherein the data interface is connected to internal bus 2, and

[0202] wherein the quantum process-based generator 15 is connected to internal bus 2, and

[0203] wherein microcontroller core 16 is connected to internal bus 2, and

[0204] wherein the quantum process-based generator 15 generates a random number at the request of microcontroller core 16, and

[0205] wherein microcontroller core 16 generates a key using of a program from one or more of its memory elements and using the random number, and

[0206] wherein microcontroller core 16 uses a program from one or more of its memory elements and the key to encrypt and decrypt data which it exchanges with devices external to the secure microcontroller, and

[0207] wherein the semiconductor substrate integrally comprises said sub-devices of the secure microcontroller 1,

[0208] wherein said sub-devices of the secure microcontroller 1 include the memory elements, internal bus 2, the at least one 8/16/32/64-bit microcontroller core 16, the data interfaces and the quantum process-based generator 15.

[0209] Item 2) The secure microcontroller 1 according to item 1),

[0210] wherein the memory elements comprise one or more read/write memories RAM 3 and/or one or more writable non-volatile memories, optionally EEPROM memories 4 and/or flash memories 4 and/or OTP memories 4, and/or one or more read-only memories and/or one or more non-volatile manufacturer memories, optionally one or more manufacturer ROMs 6, and/or one or more manufacturer EEPROMs and/or one or more manufacturer flash memories.

[0211] Item 3) The secure microcontroller 1 according to item 2),

[0212] wherein the manufacturer ROM 6 comprises the boot software.

[0213] Item 4) The secure microcontroller 1 according to item 2) or 3),

[0214] wherein manufacturer firewall 8 is provided between manufacturer memory 6 and internal bus 2.

[0215] Item 5) The secure microcontroller 1 according to one or more of items 1) to 4),

[0216] comprising one or more of the following components:

[0217] a base clock generator 21 (CLK),

[0218] a clock generator circuit 12, and/or

[0219] a reset circuit 22, and/or

[0220] a power supply or a Vcc circuit 23 with voltage regulators that provide the operating voltages, and/or

[0221] a ground circuit 24, and/or

[0222] an input/output circuit 25, and/or

[0223] one or more processing modules,

[0224] wherein the processing modules communicate with internal bus 2, and

- [0225] wherein the processing modules comprise one or more of the following modules:
- [0226] a CRC module (cyclic redundancy check) 11,
- [0227] a clock generator module 12,
- [0228] comprising a DES accelerator and/or an AES accelerator 7,
- [0229] one or more timer modules 13,
- [0230] a safety monitoring and control circuit 14,
- [0231] a data interface, optionally a universal asynchronous receiver transmitter (UART) 17.
- [0232] Item 6) The secure microcontroller 1 according to one or more of items 1) to 5),
 - [0233] comprising at least one first SPAD diode 44, and
 - [0234] comprising at least one second SPAD diode 45, and
 - [0235] comprising at least one optical fiber 50, and
 - [0236] comprising at least one processing circuit, and
 - [0237] comprising at least one operating circuit,
 - [0238] wherein the quantum process-based generator 15 comprises at least the first SPAD diode 44 as a light source for an optical quantum signal, and
 - [0239] wherein the quantum process-based generator 15 comprises at least the second SPAD diode 45 as a photodetector for the optical quantum signal, and
 - [0240] wherein the quantum process-based generator 15 comprises at least the processing circuit, and
 - [0241] wherein the quantum process-based generator 15 comprises at least the optical fiber, and
 - [0242] wherein the at least one optical fiber 50 optically couples the at least one first SPAD diode 44 with the at least one second SPAD diode 45, and
 - [0243] wherein the operating circuit supplies the first SPAD diode 44 with electrical energy such that the first SPAD diode emits light 44, and
 - [0244] wherein the processing circuit detects the signal of the second SPAD diode 45 and forms the random number therefrom and provides the latter to microcontroller core 16.
- [0245] Item 7) The secure microcontroller 1 according to item 6),
 - [0246] wherein the semiconductor substrate comprises a surface 46, and
 - [0247] wherein the semiconductor substrate comprises a semiconducting material below its surface 46, and
 - [0248] wherein the surface 46 of the semiconductor substrate comprises a metallization stack, and
 - [0249] wherein the metallization stack comprises a typically structured and optically transparent and electrically insulating layer 34, and
 - [0250] wherein at least a part of said typically structured, transparent and electrically insulating layer of surface 46 forms optical fiber 50, and
 - [0251] wherein the first SPAD diode 44 radiates from the semiconducting material of the semiconductor substrate into said optical fiber 50, and
 - [0252] wherein the optical fiber irradiates the second SPAD diode 44 in such a way that the light from inside the optical fiber 50 re-enters the semiconducting material of the semiconductor substrate from the surface and hits device parts of the second SPAD diode 44 there.
- [0253] Item 8) The secure microcontroller 1 according to item 6) and/or 7),
 - [0254] wherein the at least one operating circuit at least temporarily supplies the at least one first SPAD diode 44 with electrical energy, and
 - [0255] wherein the at least one first SPAD diode 44 feeds photons into the at least one optical fiber 50 when supplied with sufficient electrical energy, and
 - [0256] wherein the at least one optical fiber 50 emits such photons to the second SPAD diode 45.
- [0257] Item 9) The secure microcontroller 1 according to one or more of the preceding items,
 - [0258] wherein a data interface of the one or more data interfaces is a wired automotive data bus interface, and
 - [0259] wherein the wired automotive data bus interface optionally comprises
 - [0260] a CAN data bus interface, and/or
 - [0261] a CAN FD data bus interface, and/or
 - [0262] a Flexray data bus interface, and/or
 - [0263] an PSI5 data bus interface, and/or
 - [0264] an DSI3 data bus interface, and/or
 - [0265] a LIN data bus interface, and/or
 - [0266] an Ethernet data bus interface, and/or
 - [0267] a MELIBUS data bus interface.
- [0268] Item 10) The secure microcontroller 1 according to one or more of the preceding items,
 - [0269] wherein a data interface of the one or more data interfaces is a wireless data bus interface, and
 - [0270] wherein the wireless data bus interface optionally comprises
 - [0271] a WLAN interface, and/or
 - [0272] a Bluetooth interface.
- [0273] Item 11) The secure microcontroller 1 according to one or more of the preceding items,
 - [0274] wherein a data interface of the one or more data interfaces is a wired data bus interface, and
 - [0275] wherein the wireless data bus interface optionally comprises
 - [0276] a KNX data bus interface, and/or
 - [0277] an EIB data bus interface, and/or
 - [0278] a DALI data bus interface, and/or
 - [0279] a PROFIBUS data bus interface.
- [0280] Item 12) A device
 - [0281] wherein the device comprises an integrated circuit 4 with a first processor 10-1 and a non-volatile memory 16, and
 - [0282] wherein the device comprises a first memory,
 - [0283] wherein the non-volatile memory stores at least one security code,
 - [0284] wherein the first memory stores data, and
 - [0285] wherein the data in the first memory is cryptographically protected in a first format, and
 - [0286] wherein the integrated circuit is configured to validate the data read from the first memory during a transfer of data from the first memory, and
 - [0287] wherein the device comprises a quantum random number generator 28, and
 - [0288] wherein the integrated circuit and the quantum random number generator 28 are manufactured in a semiconductor crystal, and
 - [0289] wherein the semiconductor crystal has a surface 46, and
 - [0290] wherein the semiconductor crystal comprises a semiconducting material below its surface 46, and
 - [0291] wherein the surface 46 of the semiconductor crystal comprises a metallization stack, and

- [0292] wherein the metallization stack comprises a typically structured and optically transparent and electrically insulating layer 34, and
- [0293] wherein at least a part of said typically structured, transparent and electrically insulating layer 34 of surface 46 forms optical fiber 50, and
- [0294] wherein the first SPAD diode 44 radiates photons 47 from the semiconducting material of the semiconductor substrate into said optical fiber 50, and
- [0295] wherein the at least one optical fiber 50 transports such photons 48 to the second SPAD diode 45, and
- [0296] wherein the optical fiber 50 irradiates the second SPAD diode 45 in such a way that the light 49 from inside the optical fiber 50 re-enters the semiconducting material of the semiconductor substrate from surface 46 and hits device parts of the second SPAD diode 45, and
- [0297] wherein the first SPAD diode 44 and the second SPAD diode 45 and the optical fiber 50 are part of the quantum random number generator 28.
- [0298] Item 13) The device according to item 12),
- [0299] wherein the device comprises at least one operating circuit, and
- [0300] wherein the at least one operating circuit at least temporarily supplies the at least one first SPAD diode 44 with electrical energy, and
- [0301] wherein the at least one first SPAD diode 44 feeds photons 47 into the at least one optical fiber 50 when supplied with sufficient electrical energy, and
- [0302] wherein the at least one optical fiber 50 transports such photons 48 to the second SPAD diode 45, and
- [0303] wherein the at least one optical fiber 50 emits such photons 49 to the second SPAD diode 45.
- [0304] Item 14) The device according to item 13),
- [0305] wherein the quantum random number generator 28 comprises at least the first SPAD diode 44 as a light source for an optical quantum signal, and
- [0306] wherein the quantum random number generator 28 comprises at least the second SPAD diode 45 as a photodetector for the optical quantum signal, and
- [0307] wherein the quantum random number generator 28 comprises at least one processing circuit, and
- [0308] wherein the quantum random number generator 28 comprises at least optical fiber 50, and
- [0309] wherein the at least one optical fiber 50 optically couples the at least one first SPAD diode 44 with the at least one second SPAD diode 45, and
- [0310] wherein the operating circuit supplies the first SPAD diode 44 with electrical energy such that the first SPAD diode emits light 44, and
- [0311] wherein the processing circuit detects the signal of the second SPAD diode 45 and forms the random number therefrom and provides the latter to data processor 10 or another device part.
- [0312] Item 15) The device according to any one of items 12) to 14),
- [0313] wherein the first memory is external to the integrated circuit, and
- [0314] wherein the device comprises a second memory for storing data, and
- [0315] wherein the first memory is external to the integrated circuit;
- [0316] wherein the device is configured to
- [0317] transfer data from the first memory via the integrated circuit to the second memory for access by the data processor from the second memory, and
- [0318] wherein the integrated circuit is configured to
- [0319] validate the data read from the first memory during a transfer of data from the first memory to the second memory using a security code stored in the non-volatile memory, and
- [0320] if the data are validated, to apply cryptographic protection in a second format to the validated data using a security code stored in the non-volatile memory, and
- [0321] store the data protected in the second format in the second memory.
- [0322] Item 16) The device according to any one of items 12) to 15), wherein the first memory comprises a read-only memory.
- [0323] Item 17) The device according to any one of items 15) to 16), wherein the second memory comprises a random access memory.
- [0324] Item 18) The device according to any one of items 15) to 17), wherein the cryptographic protection applied to the data in the first memory is different from the cryptographic protection applied to the data in the second memory.
- [0325] Item 19) The device according to any one of items 12) to 18),
- [0326] wherein the integrated circuit includes a memory for storing data that are to be processed by the data processor, and
- [0327] wherein the device is configured to store some data of the validated data set in the memory and the rest in the second memory.
- [0328] Item 20) The device according to any one of items 15) to 19), wherein the first memory stores data in a first data format and the second memory is arranged such that it stores data in a second, different data format.
- [0329] Item 21) The device according to item 20),
- [0330] wherein the data stored in the first memory are protected by a first authentication technique, and
- [0331] wherein the device is configured to protect the data in the second memory by a second, different authentication technique.
- [0332] Item 22) The device according to any one of items 15) to 21),
- [0333] wherein the data in the first memory are stored in at least one data set and the or each data set is cryptographically protected as a set, and
- [0334] wherein the device is configured to store words or word groups of a validated data set in the second memory, wherein each word or word group is separately cryptographically protected.
- [0335] Item 23) The device according to item 22) configured to
- [0336] read the words or words groups from the second memory, and
- [0337] validates the read words or words groups by using a security code stored in the non-volatile memory, and
- [0338] to process the read and validated word or word groups in the data processor.

[0339] Item 24) The device according to item 23),

[0340] wherein the integrated circuit comprises a hash computer, and

[0341] wherein the data processor and the hash computer are arranged such that they

[0342] a) calculate a hash function for each word or each group of words depending on a security code stored in the non-volatile memory and store the hash in connection with the word or the group in the second memory,

[0343] b) retrieve a stored word or a group from the second memory, recalculate a hash function for the retrieved word or the retrieved group by using the security code, and compare the newly calculated hash with the stored hash, and

[0344] c) only allow the retrieved word or group to be processed by the data processing system if the newly calculated and stored hashes are in a certain relationship to each other.

[0345] Item 25) The device according to item 24), wherein the hash calculator is a circuit in the integrated circuit.

[0346] Item 26) The device according to any one of items 12) to 25), wherein the non-volatile memory of the integrated circuit is a one-time programmable memory.

[0347] Item 27) The device according to any one of items 12) to 26), wherein the or each data set stored in the first memory is cryptographically protected by a corresponding digital signature.

[0348] Item 28) The device according to any one of items 12) to 28), wherein the or each data set stored in the first memory is cryptographically protected by a corresponding digital signature with the aid of at least one random number of the quantum random number generator.

[0349] Item 29) The device according to item 27) or 28), wherein a security code is stored in the non-volatile memory of the integrated circuit, which security code has been generated by the device at least partially by means of at least one random number of the quantum random number generator (28).

[0350] Item 30) The device according to any one of items 27) to 29), wherein the device is configured to validate a digital signature of the data set by reference to a or the security code stored in the non-volatile memory of the integrated circuit.

[0351] Item 31) A data processing device,

[0352] wherein the data processing device comprises an integrated circuit, and

[0353] wherein the integrated circuit comprises a data processor, and

[0354] wherein the integrated circuit comprises a non-volatile memory, and

[0355] where the non-volatile memory stores at least one security code, and

[0356] wherein the integrated circuit comprises a hash computer, and

[0357] wherein the integrated circuit has an interface at the boundary of the integrated circuit, and

[0358] wherein the integrated circuit comprises a quantum random number generator, and

[0359] wherein the integrated circuit and the quantum random number generator are manufactured in a semiconductor crystal, and

[0360] wherein the semiconductor crystal has a surface 46, and

[0361] wherein the semiconductor crystal comprises a semiconducting material below its surface 46, and

[0362] wherein the surface 46 of the semiconductor crystal comprises a metallization stack, and

[0363] wherein the metallization stack comprises a typically structured and optically transparent and electrically insulating layer 34, and

[0364] wherein at least a part of said typically structured, transparent and electrically insulating layer 34 of surface 36 forms optical fiber 50, and

[0365] wherein the first SPAD diode 44 radiates photons 47 from the semiconducting material of the semiconductor substrate into said optical fiber 50, and

[0366] wherein the at least one optical fiber 50 transports such photons 48 to the second SPAD diode 45, and

[0367] wherein the optical fiber 50 irradiates the second SPAD diode 45 in such a way that the light 49 from inside the optical fiber 50 re-enters the semiconducting material of the semiconductor substrate from surface 46 and hits device parts of the second SPAD diode 45, and

[0368] wherein the first SPAD diode 44 and the second SPAD diode 45 and the optical fiber 50 are part of the quantum random number generator 28.

[0369] Item 32) A data processing device according to item 31), wherein the data processor and/or another device part of the data processing device encrypts or decrypts data with the aid of at least one random number of the quantum random number generator.

[0370] Item 33) The data process device according to items 31) or 32),

[0371] wherein the data processing device comprises a memory, and

[0372] wherein the memory is adapted to store data when used by the processor; and

[0373] wherein the memory is coupled to the data processor to receive words from the data processor and deliver words to the data processor.

[0374] Item 34) The data processing device according to any one of items 31) to 33),

[0375] wherein the memory is external to the integrated circuit, and

[0376] wherein the memory is coupled to the data processor via the interface at the boundary of the integrated circuit to receive words from the data processor and deliver words to the data processor.

[0377] Item 35) The data processing device according to any one of items 31) to 34),

[0378] wherein the data processor and the hash computer are arranged such that they

[0379] a) calculate a hash function for each word depending on a security code stored in the non-volatile memory and store the hash in connection with the word,

[0380] b) retrieve stored words from memory, recalculate a hash function for each retrieved word using the security code, and compare the recalculated hash value with the stored hash value, and

[0381] c) only permit the processing of the retrieved word by the data processing system if the newly calculated and stored hashes have a previously defined relationship.

[0382] Item 36) A device comprising the following:

[0383] an integrated circuit including a data processing means and a non-volatile memory means storing at least one security code;

[0384] a first means storing data, wherein the data is cryptographically protected in a first format by at least one authentication code; and

[0385] a quantum random number generator **28** as part of the integrated circuit,

[0386] wherein the quantum random number generator comprises a first SPAD diode **44** and a second SPAD diode **45** which are or can be coupled to each other via an optical fiber **50** fabricated outside the semiconductor substrate of the integrated circuit on the surface of the integrated circuit, and

[0387] wherein the device at least temporarily makes use of at least one random number of the quantum random number generator **28** for encrypting or decrypting a date or the authentication code.

[0388] Item 37) The device according to item 36),

[0389] wherein the device comprises a second apparatus, optionally external to the integrated circuit, for storing data, and

[0390] wherein the device comprises means for transferring data from the first memory via the integrated circuit to the second memory for access by the data processor from the second memory, and

[0391] wherein the device comprises means for validating the data read from the first memory during transfer by using a security code stored in the non-volatile memory, and

[0392] wherein the apparatus comprises means for applying cryptographic protection comprising at least one authentication code to the validated data in a second format using a security code stored in the non-volatile memory when the data is validated, and

[0393] wherein the device comprises means for storing the protected data in the second memory in the second format.

[0394] Item 38) The device, in particular according to any one of items 12) to 37),

[0395] wherein the device comprises a quantum random number generator **400**, and

[0396] wherein the quantum random number generator comprises the following device parts:

[0397] a first SPAD diode **404.1**,

[0398] a second SPAD diode **404.3**,

[0399] an optical fiber **404.2** which optically couples the first SPAD diode **404.1** and the second SPAD diode **404.3** to one another,

[0400] an amplifier **403** and/or filter,

[0401] an analog-to-digital converter **403**,

[0402] a comparator **404.2**,

[0403] a time-to-digital converter **404.3**,

[0404] an entropy extraction device **404.4** converting output values of the time-to-digital converter **403** into first and second values and generates random bits therefrom.

[0405] Item 39) The device according to item 27), wherein the device comprises a watchdog **404.5** that monitors device parts of the quantum random number generator **400**.

[0406] Item 40) The device according to any one of items 38) and 39), wherein the device comprises a voltage monitor **413** that detects and monitors analog values of analog signals.

[0407] Item 41) The device according to any one of items 38) to 40), wherein the device comprises a pseudo-random number generator **404.6**, optionally in the form of a linear feedback shift register **404.6**.

[0408] Item 42) The device according to any one of items 38) to 41), wherein the device comprises a signal multiplexer which, in the event of an error, switches from the signal of the output **411** of the entropy extraction device to a signal of an equivalent random number generator or an equivalent pseudo-random number generator **404.6**.

[0409] Item 43) The device according to any one of items 38) to 42), wherein the start value of the pseudo-random number generator **404.6** in the event of an error depends on previously correctly generated random bits of the quantum random number generator **400**.

[0410] Item 44) A method for generating a random bit comprising the steps of

[0411] generating a pulse sequence with random intervals by means of at least two SPAD diodes,

[0412] wherein the pulse sequence comprises pulses of a first height class **601** and a second height class **602**;

[0413] separating the pulses of the first height class **601** from the pulses of the second height class **602** by means of a cutting level **603**, **404.1**;

[0414] detecting **501** a first value of the time interval between a first pulse of the second height class **602** and a second pulse of the second height class **602** different from the first pulse;

[0415] detecting **501** a second value of the time interval between a third pulse of the second height class **602** different from the first pulse and a fourth pulse of the second height class **602** different from the first pulse and from the second pulse and from the third pulse,

[0416] comparing **502** the first value with the second value, and

[0417] outputting **503** a first logical value as a random bit if the first value is larger than the second value, and

[0418] outputting **503** a second logical value different from the first logical value as the random bit if the first value is smaller than the second value.

[0419] Item 45) A method **3700** for generating a quantum random number QZ with m random bits comprising the following steps:

[0420] generating **3710** a random single photon current **47**, **48**, **49**, **401.2** from single photons by means of one or more first SPAD diodes **401.1**, **44**;

[0421] transmitting **3720** the random single photon current **47**, **48**, **49**, **401.2** by means of an optical fiber **50**, **401.2** different from the semiconductor substrate **39**, **38** to one or more second SPAD diodes **401.3**, **45**;

[0422] converting **3730** of the random single photon current **47**, **48**, **49**, **401.2** into a detection signal by means of the one or more second SPAD diodes **401.3**, **45**;

[0423] conditioning **3740** the detection signal into a conditioned detection signal;

[0424] separating **3750** the pulses of the conditioned detection signal produced by coupling the emissions of a first SPAD diode **401.1** of the one or more first SPAD diodes **401.1**, **44** and a second SPAD diode **401.3**, **45** of

- the one or more second SPAD diodes **401.3, 45** from the pulses of the conditioned detection signal generated by spontaneous emission of the second SPAD diode **401.3, 45** by comparing the conditioned detection signal with a threshold value **404.1**;
- [0425] determining **3760** a first time interval between the first pulse and the second pulse of a first pair of two successive pulses of the conditioned detection signal produced by coupling the emissions of a first SPAD diode **401.1, 44** and a second SPAD diode **401.3, 45**, and determining a second time interval between a third pulse and a fourth pulse of a second pair of two successive pulses of the conditioned detection signal produced by coupling the emissions of a first SPAD diode **401.1, 44** and a second SPAD diode **401.3, 45**;
- [0426] determining **3670** the bit value of a random bit by comparing the value of the first time interval and the value of the second time interval;
- [0427] if the number **3680 n** of the determined random bits is smaller than the desired number **m** of the random bits of the quantum random number QZ to be generated, repeating the above steps **3710** to **3770** and terminating the process for generating a quantum random number if the number **3680 n** of the determined random bits is greater than or equal to the desired number **m** of the random bits of the quantum random number QZ to be generated.
- [0428] Item 46) A device comprising the following:
- [0429] an integrated circuit including a data processing means and a non-volatile memory means storing at least one security code;
- [0430] a first means storing data, wherein the data is cryptographically protected in a first format by at least one authentication code; and
- [0431] a quantum random number generator **28** as part of the integrated circuit,
- [0432] wherein the quantum random number generator comprises a first SPAD diode **44** and a second SPAD diode **45** which are or can be coupled to each other via an optical fiber **50** fabricated outside the semiconductor substrate of the integrated circuit on the surface of the integrated circuit, and
- [0433] wherein the device at least temporarily makes use of at least one random number of the quantum random number generator **28** for encrypting or decrypting a date or the authentication code.
- [0434] Item 47) The device according to item 46,
- [0435] wherein the device comprises a second apparatus, optionally external to the integrated circuit, for storing data, and
- [0436] wherein the device comprises means for transferring data from the first memory via the integrated circuit to the second memory for access by the data processor from the second memory, and
- [0437] wherein the device comprises means for validating the data read from the first memory during transfer by using a security code stored in the non-volatile memory, and
- [0438] wherein the apparatus comprises means for applying cryptographic protection comprising at least one authentication code to the validated data in a second format using a security code stored in the non-volatile memory when the data is validated, and
- [0439] wherein the device comprises means for storing the protected data in the second memory in the second format.
- [0440] Item 48) The device according to item 46) or 47),
- [0441] wherein the device comprises a quantum random number generator **400**, and
- [0442] wherein the quantum random number generator comprises the following device parts:
- [0443] a first SPAD diode **404.1**,
- [0444] a second SPAD diode **404.3**,
- [0445] an optical fiber **404.2** which optically couples the first SPAD diode **404.1** and the second SPAD diode **404.3** to one another,
- [0446] an amplifier **403** and/or filter,
- [0447] an analog-to-digital converter **403**,
- [0448] a comparator **404.2**,
- [0449] a time-to-digital converter **404.3**,
- [0450] an entropy extraction device **404.4** converting output values of the time-to-digital converter **403** into first and second values and generates random bits for the random number therefrom.
- [0451] Item 49) The device according to item 48), wherein the device comprises a watchdog **404.5** that monitors device parts of the quantum random number generator **400**.
- [0452] Item 50) The device according to any one of items 48) and 49), wherein the device comprises a voltage monitor **413** that detects and monitors analog values of analog signals.
- [0453] Item 51) The device according to any one of items 48) to 50), wherein the device comprises a random number generator a pseudo-random number generator **404.6**, optionally in the form of a linear feedback shift register **404.6**.
- [0454] Item 52) The device according to any one of items 48) to 51), wherein the device comprises a signal multiplexer which, in the event of an error, switches from the signal of the output **411** of the entropy extraction device to a signal of an equivalent random number generator or an equivalent pseudo-random number generator **404.6**.
- [0455] Item 53) The device according to any one of items 48) to 52), wherein the start value of the pseudo-random number generator **404.6** in the event of an error depends on previously correctly generated random bits of the quantum random number generator **400**.
- [0456] Item 54) A method for generating a random bit comprising the steps of
- [0457] generating a pulse sequence with random intervals by means of at least two SPAD diodes,
- [0458] wherein the pulse sequence comprises pulses of a first height class **601** and a second height class **602**;
- [0459] separating the pulses of the first height class **601** from the pulses of the second height class **602** by means of a cutting level **603, 404.1**;
- [0460] detecting **501** a first value for the time interval between a first pulse of the second height class **602** and a second pulse of the second height class **602** different from the first pulse;
- [0461] detecting **501** a second value for the time interval between a third pulse of the second height class **602** different from the first pulse and a fourth pulse of the second height class **602** different from the first pulse and from the second pulse and from the third pulse;
- [0462] comparing **502** the first value with the second value, and

[0463] outputting 503 a first logical values as a random bit if the first value is larger than the second value, and

[0464] outputting 503 a second logical value different from the first logical value as the random bit if the first value is smaller than the second value.

[0465] Item 55) A method 3700 for generating a quantum random number QZ with m random bits comprising the following steps:

[0466] generating 3710 a random single photon current 47, 48, 49, 401.2 from single photons by means of one or more first SPAD diodes 401.1, 54;

[0467] transmitting 3720 the random single photon current 47, 48, 49, 401.2 by means of an optical fiber 50, 401.2 to one or more second SPAD diodes 401.3, 45;

[0468] converting 3730 of the random single photon current 47, 48, 49, 401.2 into a detection signal by means of the one or more second SPAD diodes 401.3, 45;

[0469] conditioning 3740 the detection signal into a conditioned detection signal;

[0470] separating 3750 the pulses of the conditioned detection signal produced by coupling the emissions of a first SPAD diode 401.1 of the one or more first SPAD diodes 401.1, 44 and a second SPAD diode 401.3, 45 of the one or more second SPAD diodes 401.3, 45 from the pulses of the conditioned detection signal generated by spontaneous emission of the second SPAD diode 401.3, 45 by comparing the conditioned detection signal with a threshold value 404.1;

[0471] determining 3760 a first time interval between the first pulse and the second pulse of a first pair of two successive pulses of the conditioned detection signal produced by coupling the emissions of a first SPAD diode 401.1, 44 and a second SPAD diode 401.3, 45, and determining a second time interval between a third pulse and a fourth pulse of a second pair of two successive pulses of the conditioned detection signal produced by coupling the emissions of a first SPAD diode 401.1, 44 and a second SPAD diode 401.3, 45;

[0472] determining 3670 the bit value of a random bit by comparing the value of the first time interval and the value of the second time interval;

[0473] if the number 3680 n of the determined random bits is smaller than the desired number m of the random bits of the quantum random number QZ to be generated, repeating the above steps 3710 to 3770 and terminating the process for generating a quantum random number if the number 3680 n of the determined random bits is greater than or equal to the desired number m of the random bits of the quantum random number QZ to be generated.

[0474] 56) A device for generating a quantum random number QZ,

[0475] comprising one or more first SPAD diodes 401.1, 44 generating an optically quantum process-based random single photon current 47, 48, 49, 401.2 from single photons 3710, and

[0476] comprising one or more second SPAD diodes 401.3, 45, and

[0477] comprising an optical fiber 50, 401.2 different from the semiconductor substrate 39, 38, which transmits the random single photon current 47, 48, 49, 401.2 to one or more second SPAD diodes 401.3, 45 3720, and

[0478] wherein the one or more second SPAD diodes 401.3, 45 convert the random single photon current 47, 48, 49, 401.2 into a detection signal 3730, and

[0479] wherein a signal processing device, optionally an amplifier 402, conditions the detections signals into a conditioned detections signal 3740, and

[0480] wherein a comparators 404.02 or a functionally equivalent device separates 3750 the pulses of the conditioned detection signal produced by coupling the emissions of a first SPAD diode 401.1 of the one or more first SPAD diodes 401.1, 44 and a second SPAD diode 401.3, 45 of the one or more second SPAD diodes 401.3, 45 from the pulses of the conditioned detection signal generated by spontaneous emission of the second SPAD diode 401.3, 45 by comparing the conditioned detection signal with a threshold value 404.1, and

[0481] wherein a time-to-digital converter 404.3 determines 3760 a first time interval between the first pulse and the second pulse of a first pair of two successive pulses of the conditioned detection signal produced by coupling the emissions of a first SPAD diode 401.1, 44 and a second SPAD diode 401.3, 45, and determining a second time interval between a third pulse and a fourth pulse of a second pair of two successive pulses of the conditioned detection signal produced by coupling the emissions of a first SPAD diode 401.1, 44 and a second SPAD diode 401.3, 45; and

[0482] wherein an entropy extraction device 404.4 determines 3670 the bit value of a random bit by comparing the value of the first time interval and the value of the second time interval, and

[0483] wherein a finite-state machine 404.8 generates a quantum random number QZ 417 from the bit data stream of random bits 411.

LIST OF REFERENCE NUMERALS

[0484] 1 secure microcontroller

[0485] 2 one or more internal data buses

[0486] 3 one or more read/write memories RAM

[0487] 4 one or more writable non-volatile memories; said non-volatile memories can comprise EEPROM memories or flash memories or OTP memories, for example

[0488] 5 one or more non-volatile, read-only memories, such as a ROM

[0489] 6 one or more non-volatile, writable and/or non-writable manufacturer memories; in the case on a non-writable manufacturer memory, the manufacturer memory can be a manufacturer ROM, for example

[0490] 7 one or more cryptographic accelerators, for example a DES accelerator and/or an AES accelerator

[0491] 8 one or more manufacturer memory firewalls

[0492] 11 one or more CRC modules (cyclic redundancy check)

[0493] 12 one or more clock generator modules (CLK)

[0494] 13 timer modules

[0495] 14 one or more safety monitoring and control circuits

[0496] 15 quantum process-based generator

[0497] 16 microcontroller core

[0498] 17 data interface, optionally one or more universal asynchronous receiver transmitters (UART) to support high-speed serial data

[0499] 21 one or more base clock generators (CLK)

- [0500] 22 one or more reset circuits
- [0501] 23 one or more power supply or Vcc circuits with voltage regulators that provide the
- [0502] operating voltages for the secure microcontroller
- [0503] 24 one or more ground circuits
- [0504] 25 one or more input/output circuits
- [0505] 30 exemplary SPAD diode for use as a sensor element of a single-photon detector
- [0506] 31 shallow trench isolation STI of the exemplary SPAD diode
- [0507] 32 anode contact of the exemplary SPAD diode
- [0508] 33 cathode contact of the exemplary SPAD diode; the cathode contact of the exemplary SPAD diode is preferably made of indium tin oxide (ITO) or another transparent and electrically conductive material
- [0509] 34 insulating layer
- [0510] 34' insulating layer
- [0511] 34" insulating layer
- [0512] 35 highly doped first connection area of a first line type; in a CMOS technology with a p-doped wafer material, for example, it can be an n⁺-doped area in the semiconducting substrate material of the SPAD diode
- [0513] 36 first doped tray of a second line type; in a CMOS technology with a p-doped wafer material, for example, it can be a weaker doped, i.e. p⁺-doped area in the semiconducting substrate material of the SPAD diode 1820
- [0514] 37 second doped tray of a second line type; in a CMOS technology with a p-doped wafer material, for example, it can be a weaker doped, i.e. p⁺-doped area in the semiconducting substrate material of the SPAD diode
- [0515] 38 epitaxial layer of a second line type; in a CMOS technology with a p-doped wafer material, for example, it can be a p-doped epitaxial layer in the semiconducting substrate material of the SPAD diode
- [0516] 39 base material of the semiconducting monocrystalline wafer, which has a second conductivity type; in a CMOS technology with a p-doped wafer material, for example, it is a p-doped monocrystalline semiconductor wafer
- [0517] 40 second doped tray of a second line type below the anode contact; in a CMOS technology with a p-doped wafer material, for example, it can be a p⁺-doped area in the semiconducting substrate material of the SPAD diode
- [0518] 41 highly doped second connection area of a second line type; in a CMOS technology with a p-doped wafer material, for example, it can be a p⁺-doped area in the semiconducting substrate material of the SPAD 1820 diode
- [0519] 42 insulation, for example an oxide or similar
- [0520] 43 metal cover of the optical fiber
- [0521] 44 first SPAD diode; the first SPAD diode serves at least temporarily as a light source for irradiating the second SPAD diode with photons from the first SPAD diode
- [0522] 45 second SPAD diode; the second SPAD diode serves, for example, at least temporarily as a photodetector for the light from the first SPAD diode
- [0523] 46 surface of wafer in line with the application submitted herein
- [0524] 47 light of the first SPAD diode emitted vertically upwards in a direction perpendicular to the surface
- [0525] 48 light transported horizontally in the optical fiber, which is part of the light emitted vertically into the optical fiber by the first SPAD diode
- [0526] 49 light of the first SPAD diode emitted vertically downwards in a direction perpendicular to the surface from the optical fiber into the second SPAD diode, which light was emitted by the first SPAD diode as a perpendicular light into the optical fiber and then transported horizontally from the optical fiber to the second SPAD diode;
- [0527] 50 optical fiber for transporting the photons from the first SPAD diode to the second SPAD diode; the optical fiber is formed from a covering oxide or another optically transparent insulating layer on the circuit of the exemplary SPAD diode.
- [0528] 50' optical fiber for transporting the photons from the first SPAD diode to the second SPAD diode; the optical fiber is formed from a covering oxide or another optically transparent insulating layer on the circuit of the exemplary SPAD diode or from two optically transparent insulating layers (e.g. the metalization stack) arranged on top of each other.
- [0529] 51 contacting
- [0530] 52 contacting
- [0531] 53 plane between two electrically insulating layers of the waveguide
- [0532] 400 quantum random number generator QRNG
- [0533] 401 entropy source
- [0534] 401.1 one or more first SPAD diodes
- [0535] 401.2 optical fiber
- [0536] 401.3 one or more second SPAD diodes
- [0537] 402 high-frequency amplifier
- [0538] 403 analog-to-digital converter (ADC)
- [0539] 404 measurement board with FPGA
- [0540] 404.1 constant
- [0541] 404.2 comparator
- [0542] 404.3 time-to-digital converter
- [0543] 404.4 entropy extraction device
- [0544] 404.5 watchdog
- [0545] 404.6 linear feedback shift register; the feedback is preferably a simple primitive polynomial to generate pseudo-random bit sequences;
- [0546] 404.7 signal multiplexer
- [0547] 404.8 finite-state machine
- [0548] 404.9 RAM
- [0549] 404.10 Finish Flag
- [0550] 404.11 microcontroller
- [0551] 405 voltage signal of the entropy source 401
- [0552] 406 amplifier output signal 406 of the high-frequency amplifier 402
- [0553] 407 digital 14-bit value 407 of the analog-to-digital converter 403; other bit widths are conceivable
- [0554] 408 signal of the constant 404.1
- [0555] 409 output signal 409 of the comparator 404.2
- [0556] 410 output 410 of the time-to-digital converter 404.3
- [0557] 411 output of the entropy extraction 404.4
- [0558] 412 seed S
- [0559] 413 voltage monitor
- [0560] 414 signal lines
- [0561] 416 selection signal

[0562] 417 pseudo-random signal line
 [0563] 418 random data words
 [0564] 419 internal data bus of quantum random number generator 400; preferably, this is the internal data bus of control device 4
 [0565] 420 interrupt signal of watchdog 404.5 of quantum random number generator 400 and control device 4 of fuse 1, respectively
 [0566] 500 flow diagram 500 of entropy extraction method
 [0567] 501 first step 501 with determining the first value of the output 410 of the time-to-digital converter 404.3 and the second value of the output 410 of the time-to-digital converter
 [0568] 404.3 and storing in a shift register of the entropy extraction 404.4
 [0569] 502 second step of comparing the first value with the second value
 [0570] 503 third step of evaluating the first value and the second value and generating the random bit
 [0571] 601 first spikes
 [0572] 602 second spikes
 [0573] 603 cutting level
 [0574] 3700 method for generating a quantum random number QZ with m random bits
 [0575] 3710 generating a random single photon current (57, 58, 59, 401.2) by means of one or more first SPAD diodes (401.1, 54)
 [0576] 3720 transmitting the random single photon current (57, 58, 59, 401.2) by means of an optical fiber (44, 401.2) different from the semiconductor substrate (49, 48) to one or more second SPAD diodes (401.3, 55)
 [0577] 3730 converting the random single photon current (57, 58, 59, 401.2) into a detection signal in the form of a voltage signal 405 of the entropy source 401, which preferably comprises the first SPAD diodes 401.1 and the optical fiber 401.2 and the second SPAD diodes 401.3
 [0578] 3740 conditioning, optionally amplifying and/or filtering and/or analog-to-digital converting, of the detection signal into a conditioned detection signal, optionally a digital 14-bit value 407 of the analog-to-digital converter 403
 [0579] 3750 separating the pulses of the conditioned detection signal produced by coupling the emissions of a first SPAD diode 401.1 and a second SPAD diode 401.3 from the pulses of the conditioned detection signal produced by spontaneous emission by comparing the conditioned detection signal with a threshold value, optionally in a comparator 404.2, and generating a corresponding output signal 409, optionally of the comparator 404.2
 [0580] 3760 determining a first time interval between the first pulse and the second pulse of a first pair of two successive pulses of the conditioned detection signal produced by coupling the emissions of a first SPAD diode 401.1 and a second SPAD diode 401.3, and determining a second time interval between a third pulse and a fourth pulse of a second pair of two successive pulses of the conditioned detection signal produced by coupling the emissions of a first SPAD diode 401.1 and a second SPAD diode 401.3 and, optionally, to determine the first value of the output 410

of the time-to-digital converter 404.3 and the second value of the output 410 of the time-to-digital converter 404.3

[0581] 3670 determining the bit value of a random bit by comparing the value of the first time interval and the value of the second time interval

[0582] 3680 If the number n of random bits determined up to this step is smaller than the desired number m of random bits of the desired quantum random number, the above steps are repeated. Otherwise, the process is terminated to generate a quantum random number with m random bits.

What is claimed is:

1. A device, wherein the device comprises a quantum random number generator, further comprising following device parts:

a first SPAD diode,
 a second SPAD diode,
 an optical fiber which optically couples the first SPAD diode and the second SPAD diode to one another,
 an amplifier and/or filter,
 an analog-to-digital converter,
 a comparator,
 a time-to-digital converter, and
 an entropy extraction device converting output values of the time-to-digital converter into first and second values and generates random bits therefrom.

2. The device according to claim 1, wherein the device further comprises a watchdog that monitors the device parts of the quantum random number generator.

3. The device according to claim 1, wherein the device further comprises a voltage monitor arranged to detect and monitor analog values of analog signals.

4. The device according to claim 1, wherein the device further comprises a pseudo-random number generator.

5. The device according to claim 4, wherein the pseudo-random number generator is in form of a linear feedback shift register.

6. The device according to claim 4, wherein the device further comprises a signal multiplexer which, in an event of an error, switches from an output of the entropy extraction device to an equivalent random number generator or an equivalent pseudo-random number generator.

7. The device according to claim 4, wherein a start value of a pseudo-random number generator in an event of an error depends on previously correctly generated random bits of the quantum random number generator.

8. A method for generating a random bit comprising:

generating a pulse sequence with random intervals by at least two SPAD diodes, wherein the pulse sequence comprises pulses of a first height class and a second height class;

separating the pulses of the first height class from the pulses of the second height class by a cutting level;

detecting a first value of a time interval between a first pulse of the second height class and a second pulse of the second height class different from the first pulse;

detecting a second value of the time interval between a third pulse of the second height class different from the first pulse and a fourth pulse of the second height class

different from the first pulse and from the second pulse and from the third pulse,
 comparing the first value with the second value,
 outputting a first logical value as a random bit if the first value is larger than the second value, and
 outputting a second logical value different from the first logical value as the random bit if the first value is smaller than the second value.

9. A method for generating a quantum random number QZ with m random bits comprising:

generating a random single photon current from single photons by one or more first SPAD diodes;

transmitting the random single photon current by an optical fiber different from a semiconductor substrate to one or more second SPAD diodes;

converting of the random single photon current into a detection signal by the one or more second SPAD diodes;

conditioning the detection signal into a conditioned detection signal;

separating pulses of the conditioned detection signal produced by coupling emissions of a first SPAD diode of the one or more first SPAD diodes and a second SPAD diode of the one or more second SPAD diodes from the pulses of the conditioned detection signal generated by spontaneous emission of the second SPAD diode by comparing the conditioned detection signal with a threshold value;

determining a first time interval between a first pulse and a second pulse of a first pair of two successive pulses of the conditioned detection signal produced by coupling the emissions of the first SPAD diode and the second SPAD diode, and determining a second time interval between a third pulse and a fourth pulse of a second pair of two successive pulses of the conditioned detection signal produced by coupling the emissions of the first SPAD diode and the second SPAD diode;

determining a bit value of a random bit by comparing a first value of the first time interval and a second value of the second time interval; and

if a number n of the determined random bits is smaller than a desired number m of the determined random bits of the quantum random number QZ to be generated, repeating above steps and terminating this process for generating the quantum random number QZ if the number n of the determined random bits is greater than or equal to the desired number m of the determined random bits of the quantum random number QZ to be generated.

10. A device comprising:

an integrated circuit including a data processor and a non-volatile memory storing at least one security code;

a first memory, wherein data in the first memory is cryptographically protected in a first format by at least one authentication code;

a quantum random number generator as part of the integrated circuit,
 wherein the quantum random number generator comprises a first SPAD diode and a second SPAD diode which are or can be coupled to each other via an optical fiber fabricated outside a semiconductor substrate of the integrated circuit on a surface of the integrated circuit, and

wherein the device at least temporarily makes use of at least one random number of the quantum random number generator for encrypting or decrypting a date or the at least one authentication code.

11. The device according to claim 10,

wherein the device further comprises a second memory, external to the integrated circuit, for storing data,
 wherein the device is configured to transfer data from the first memory via the integrated circuit to the second memory for access by the data processor from the second memory,

wherein the device is configured to validate the data read from the first memory during transfer by using the at least one security code stored in the non-volatile memory,

wherein the device is configured to apply cryptographic protection comprising the at least one authentication code to the validated data in a second format using the at least one security code stored in the non-volatile memory when the data is validated, and

wherein the device is configured to store the protected data in the second memory in the second format.

12. The device according to claim 10,

wherein the device comprises a quantum random number generator, further comprising following device parts:

a first SPAD diode,

a second SPAD diode,

an optical fiber which optically couples the first SPAD diode and the second SPAD diode to one another,

an amplifier and/or filter,

an analog-to-digital converter,

a comparator,

a time-to-digital converter, and

an entropy extraction device converting output values of the time-to-digital converter into first and second values and generates random bits of a random number therefrom.

13. The device according to claim 12, wherein the device comprises a watchdog that monitors one or more of the device parts of the quantum random number generator.

14. The device according to claim 12, wherein the device further comprises a voltage monitor arranged to detect and monitor analog values of analog signals.

15. The device according to claim 12, wherein the device further comprises a random number generator a pseudo-random number generator.

16. The device according to claim 15, wherein the pseudo-random number generator is in form of a linear feedback shift register.

17. The device according to claim 15, wherein the device further comprises a signal multiplexer which, in an event of an error, switches from an output of the entropy extraction device to an equivalent random number generator or an equivalent pseudo-random number generator.

18. The device according to claim 15, wherein a start value of the pseudo-random number generator in an event of an error depends on previously correctly generated random bits of the quantum random number generator.

19. A method for generating a random bit comprising:

generating a pulse sequence with random intervals by use of at least two SPAD diodes, wherein the pulse sequence comprises pulses of a first height class (601) and a second height class;

separating the pulses of the first height class from the pulses of the second height class by a cutting level;
 detecting a first value for a time interval between a first pulse of the second height class and a second pulse of the second height class different from the first pulse;
 detecting a second value for the time interval between a third pulse of the second height class different from the first pulse and a fourth pulse of the second height class different from the first pulse and from the second pulse and from the third pulse;

comparing the first value with the second value,
 outputting a first logical value as a random bit if the first value is larger than the second value, and
 outputting a second logical value different from the first logical value as the random bit if the first value is smaller than the second value.

20. A method for generating a quantum random number QZ with m random bits comprising:

generating a random single photon current from single photons by use of one or more first SPAD diodes;
 transmitting the random single photon current by an optical fiber to one or more second SPAD diodes;
 converting of the random single photon current into a detection signal by use of the one or more second SPAD diodes;

conditioning the detection signal into a conditioned detection signal;

separating pulses of the conditioned detection signal produced by coupling emissions of a first SPAD diode of the one or more first SPAD diodes and a second SPAD diode of the one or more second SPAD diodes from the pulses of the conditioned detection signal generated by spontaneous emission of the second SPAD diode by comparing the conditioned detection signal with a threshold value;

determining a first time interval between a first pulse and a second pulse of a first pair of two successive pulses of the conditioned detection signal produced by coupling the emissions of a first SPAD diode and a second SPAD diode, and determining a second time interval between a third pulse and a fourth pulse of a second pair of two successive pulses of the conditioned detection signal produced by coupling the emissions of a first SPAD diode and a second SPAD diode;

determining a bit value of a random bit by comparing the value of the first time interval and the value of the second time interval; and

if a number n of the determined random bits is smaller than a desired number m of the random bits of the

quantum random number QZ to be generated, repeating above steps and terminating this process for generating the quantum random number QZ if the number n of the determined random bits is greater than or equal to the desired number m of the random bits of the quantum random number QZ to be generated.

21. A device for generating a quantum random number QZ, comprising:

one or more first SPAD diodes generating an optically quantum process-based random single photon current from single photons,

one or more second SPAD diodes,

an optical fiber different from a semiconductor substrate, which transmits the random single photon current to one or more second SPAD diodes,

wherein the one or more second SPAD diodes convert the random single photon current into a detection signal,

wherein a signal processing device including an amplifier conditions the detections signals into a conditioned detections signal,

wherein a comparator or a functionally equivalent device separates pulses of the conditioned detection signal produced by coupling emissions of a first SPAD diode of the one or more first SPAD diodes and a second SPAD diode of the one or more second SPAD diodes from the pulses of the conditioned detection signal generated by spontaneous emission of the second SPAD diode by comparing the conditioned detection signal with a threshold value,

wherein a time-to-digital converter determines a first time interval between a first pulse and a second pulse of a first pair of two successive pulses of the conditioned detection signal produced by coupling the emissions of a first SPAD diode and a second SPAD diode, and determining a second time interval between a third pulse and a fourth pulse of a second pair of two successive pulses of the conditioned detection signal produced by coupling the emissions of a first SPAD diode and a second SPAD diode;

wherein an entropy extraction device determines a bit value of a random bit by comparing a first value of the first time interval and a second value of the second time interval, and

wherein a finite-state machine generates the quantum random number QZ from a bit data stream of random bits.

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